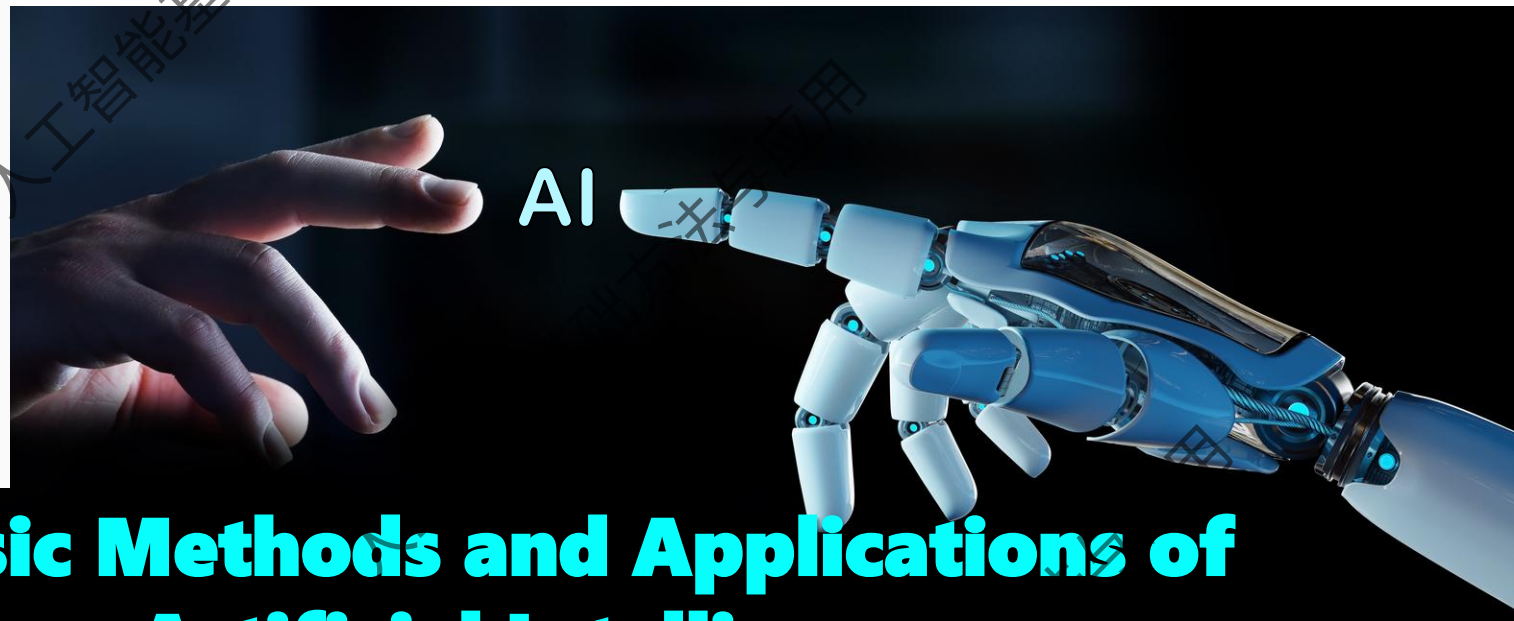
群聊：人工智能基础与应用I-06班

该二维码7天内（10月4日前）有效



Basic Methods and Applications of Artificial Intelligence

——监督学习

人工智能学院 / 人工智能与机器人研究所
人机混合增强智能全国重点实验室
陈炜煌

IAIR Est. 1986
Institute of Artificial Intelligence and Robotics, XJTU

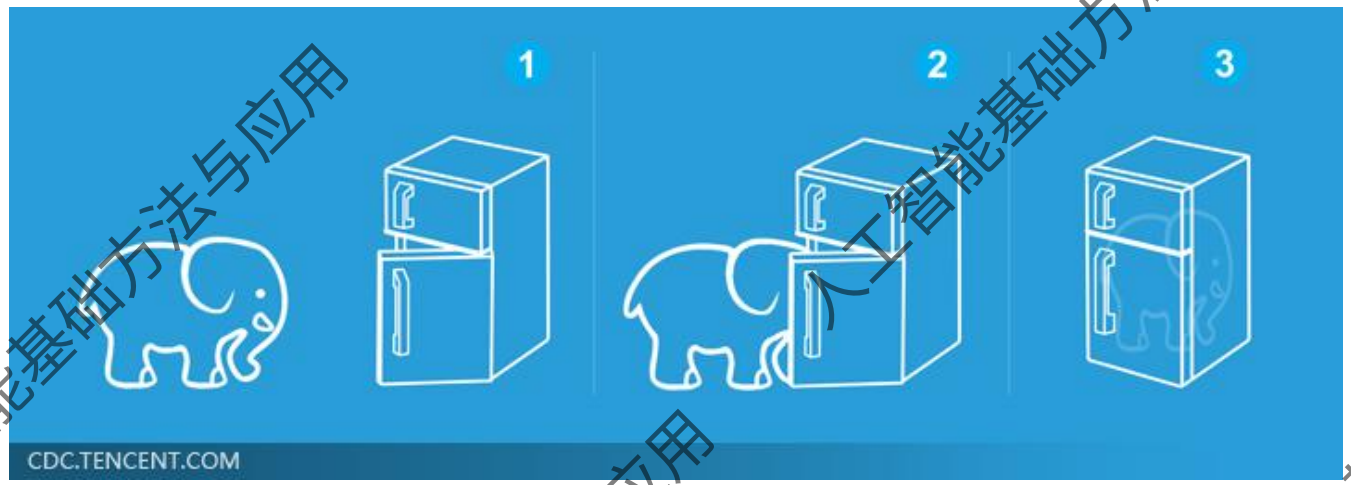
IAIR Est. 1986

Machine Learning is so simple

Step 1: function with unknown

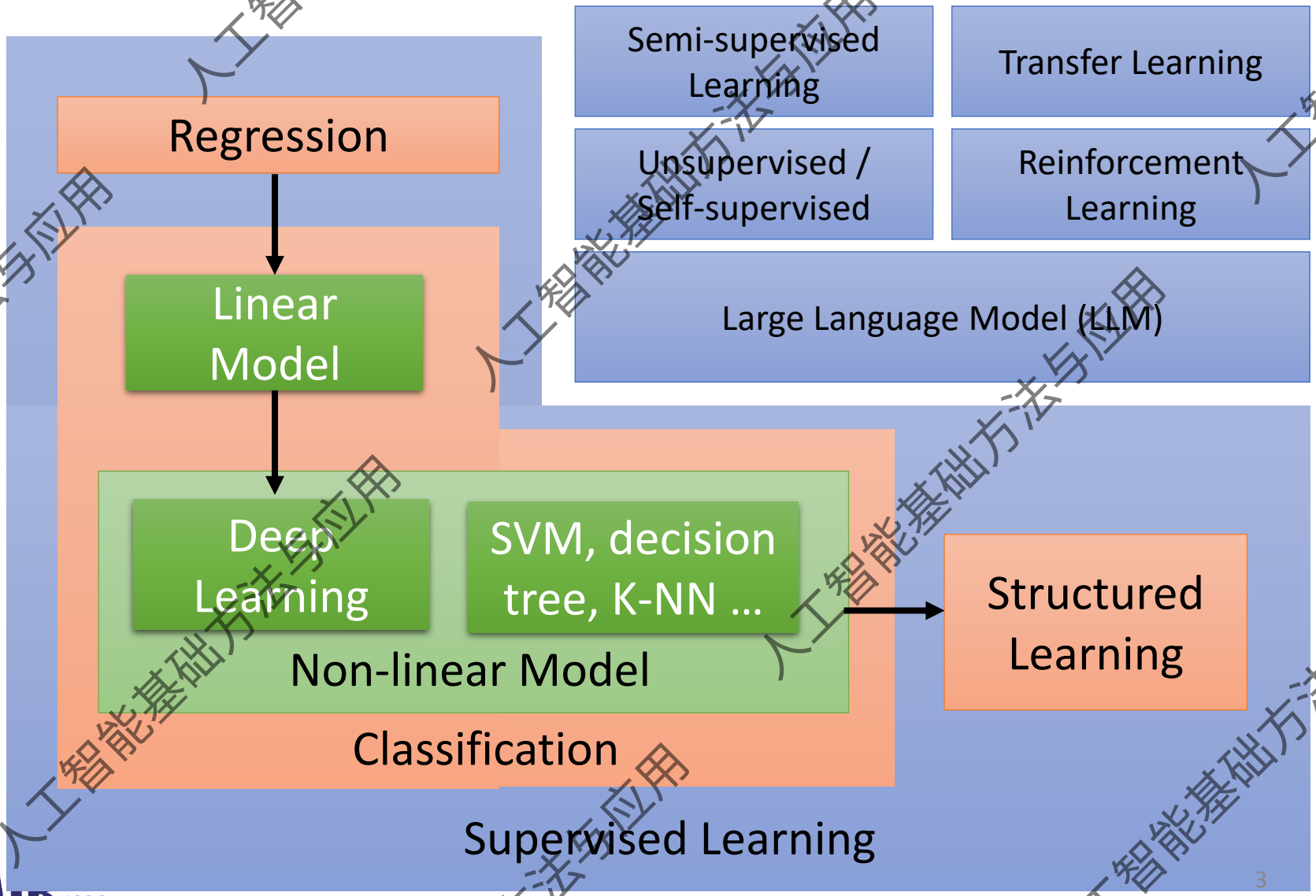
Step 2: define loss from training data

Step 3: optimization



人工智能基础方法与应用

机器学习的学习地图



人工智能基础方法与应用

一、回归

人工智能基础方法与应用

什么是回归

- 股票市场预测

$f($  $)=$ 道琼斯工业平均指数

- 自动驾驶

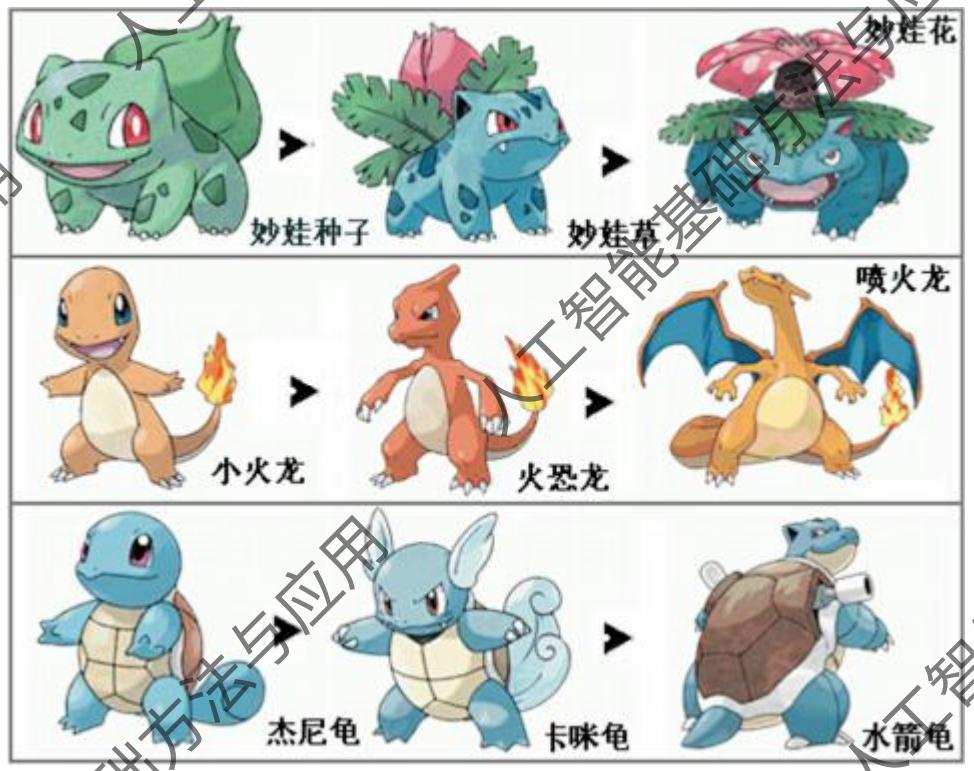
$f($  $)=$ 方向盘角度

- 推荐系统

$f($ 使用者A 商品B $)=$ 购买可能性

人工智能基础方法与应用

应用示例：宝可梦进化预测

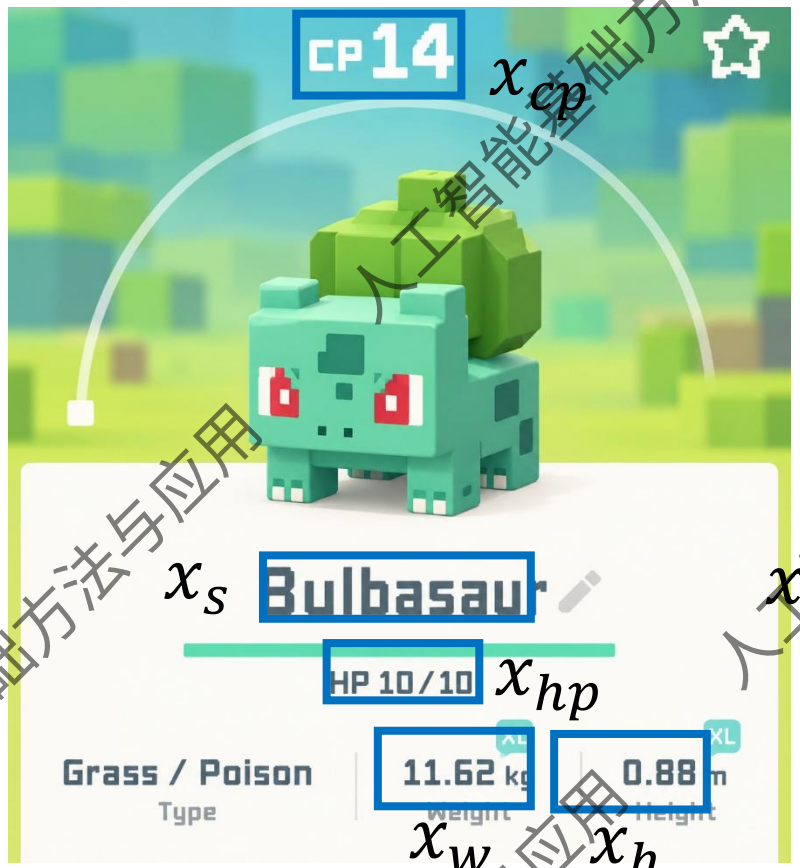




人工智能基础方法与应用

应用示例

- 预测一只宝可梦进化后的战斗力(Combat Power， CP)

$f($ 

x $) =$ CP after evolution y

Step 1: Model

$y = b + w \cdot x_{cp}$

A set of function

Model

$f_1, f_2 \cdots$

w and b are parameters (can be any value)

f_1: $y = 10.0 + 9.0 \cdot x_{cp}$

f_2: $y = 9.8 + 9.2 \cdot x_{cp}$

f_3: $y = -0.8 - 1.2 \cdot x_{cp}$

..... infinite

$f($ 

$x) =$ CP after evolution y

Linear model: $y = b + \sum w_i x_i$

x_i: an attribute of input x — feature

w_i: weight, b: bias

IAIR Est. 1986

Step 2: Goodness of Function

$y = b + w \cdot x_{cp}$

A set of function

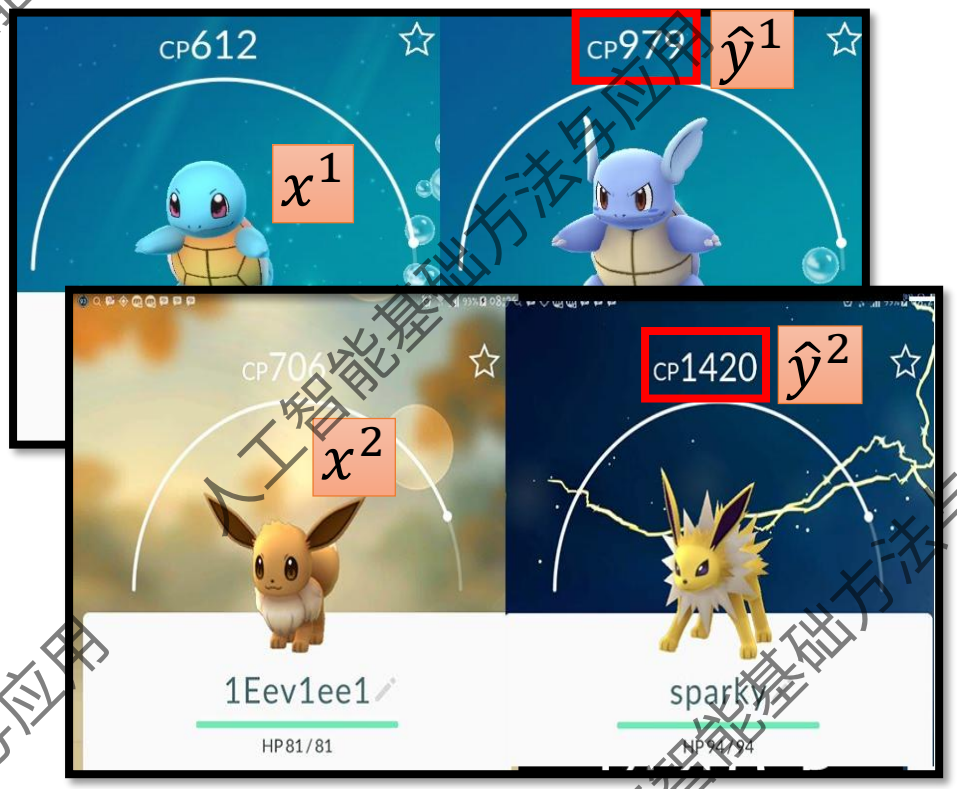
Model

$f_1, f_2 \cdots$

Training Data

function input:

function Output (scalar):



人工智能基础方法与应用

IAIR Est. 1986

Step 2: Goodness of Function

Training Data:
10 pokemons

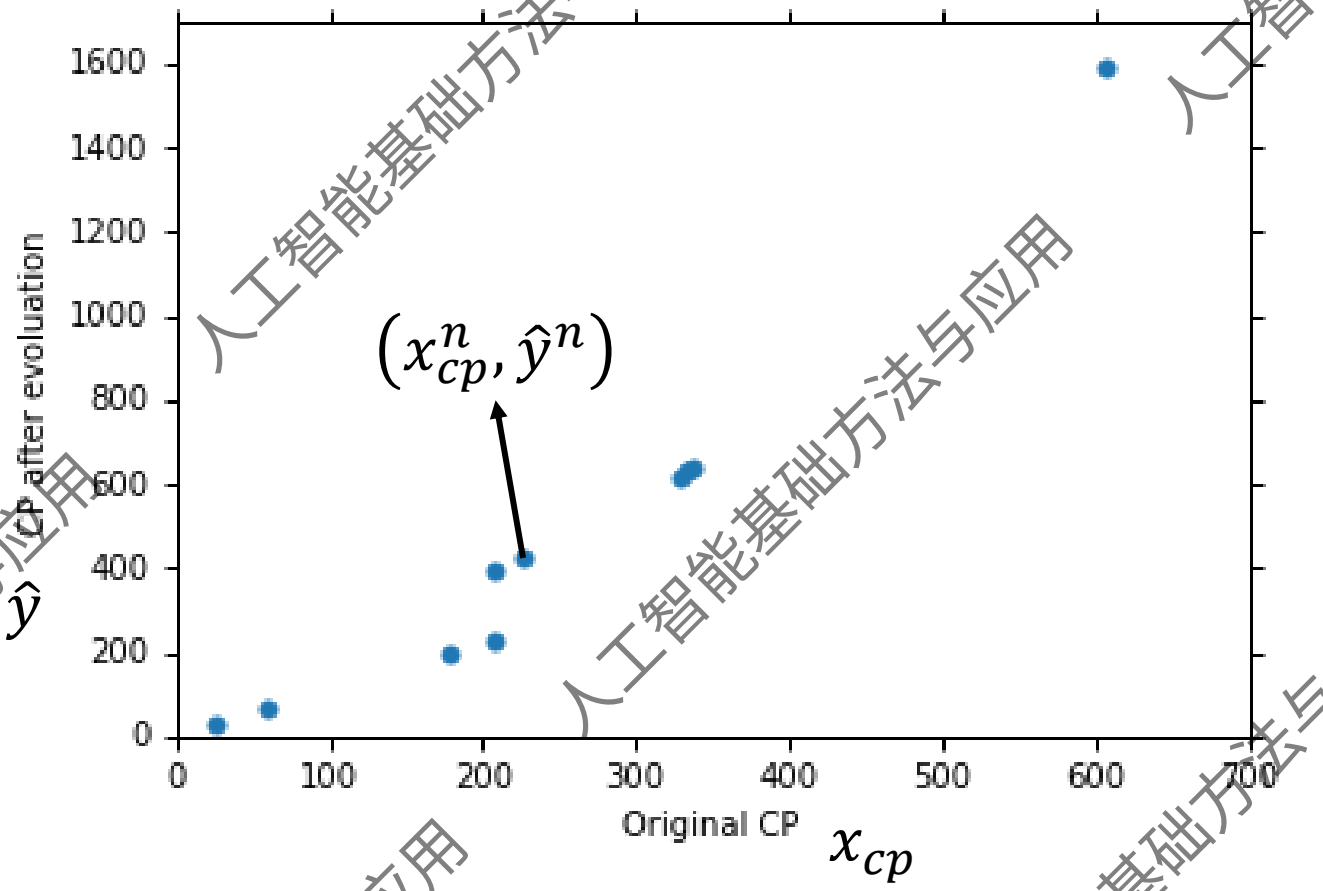
$(x^1, \hat{y}^1)$

$(x^2, \hat{y}^2)$

$\vdots$

$(x^{10}, \hat{y}^{10})$

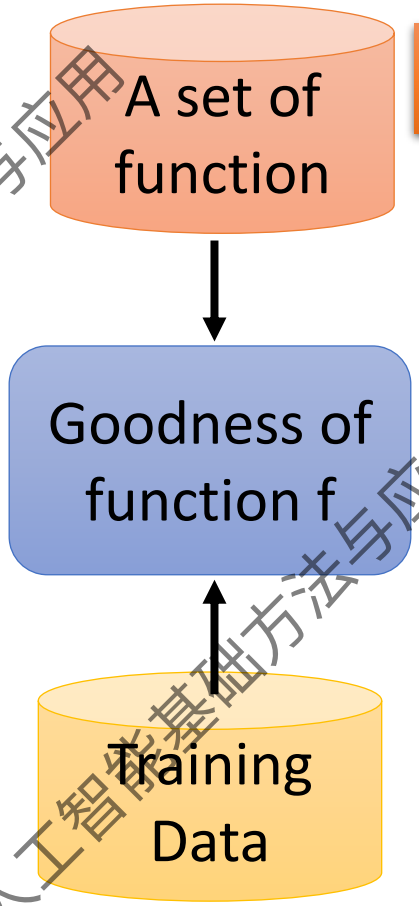
This is real data.



人工智能基础方法与应用

Step 2: Goodness of Function

$y = b + w \cdot x_{cp}$



Model

$f_1, f_2 \cdots$

Loss function L:

Input: a function, output: how bad it is

$$L(f) = L(w, b)$$

Estimated y based on input function

$$= \sum_{n=1}^{10} \left(\hat{y}^n - \left(b + w \cdot x_{cp}^n \right) \right)^2$$

Estimation error

Sum over examples

Step 2: Goodness of Function

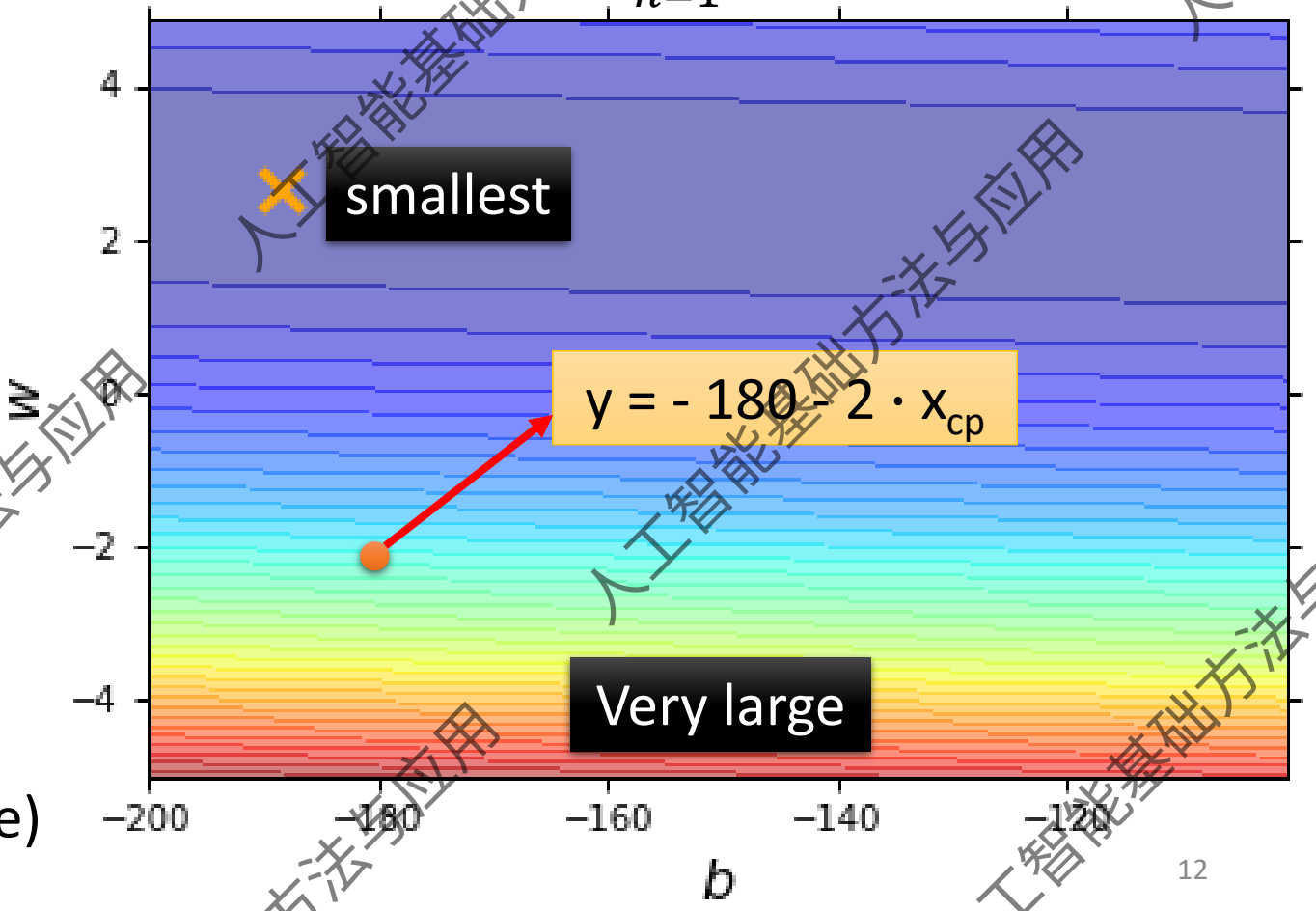
- Loss Function

$$L(w,b) = \sum_{n=1}^{10} \left(\hat{y}^n - \left(b + w \cdot x_{cp}^n\right)\right)^2$$

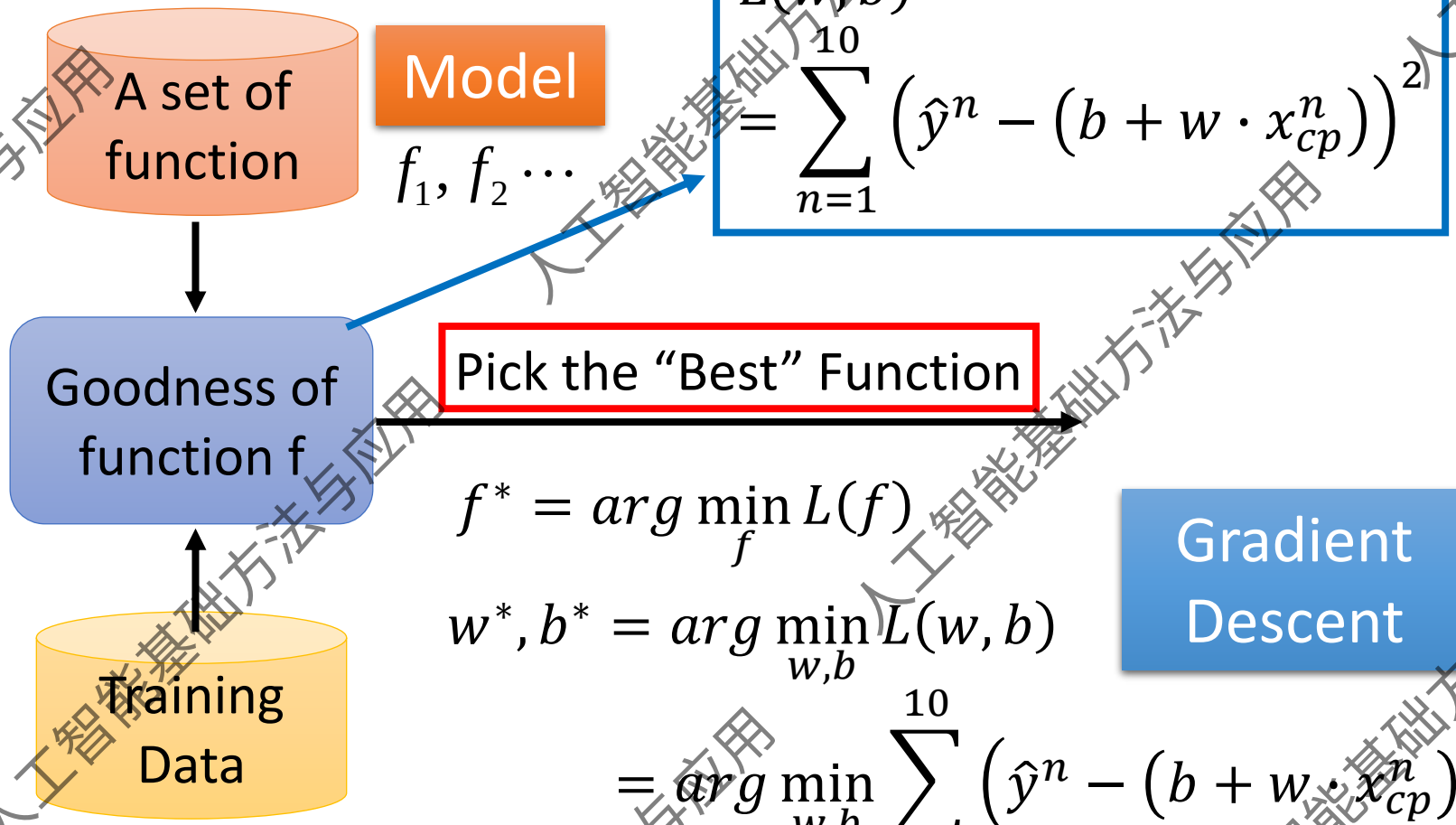
Each point in the figure is a function

The color represents $L(w,b)$.

(true example)



Step 3: Best Function

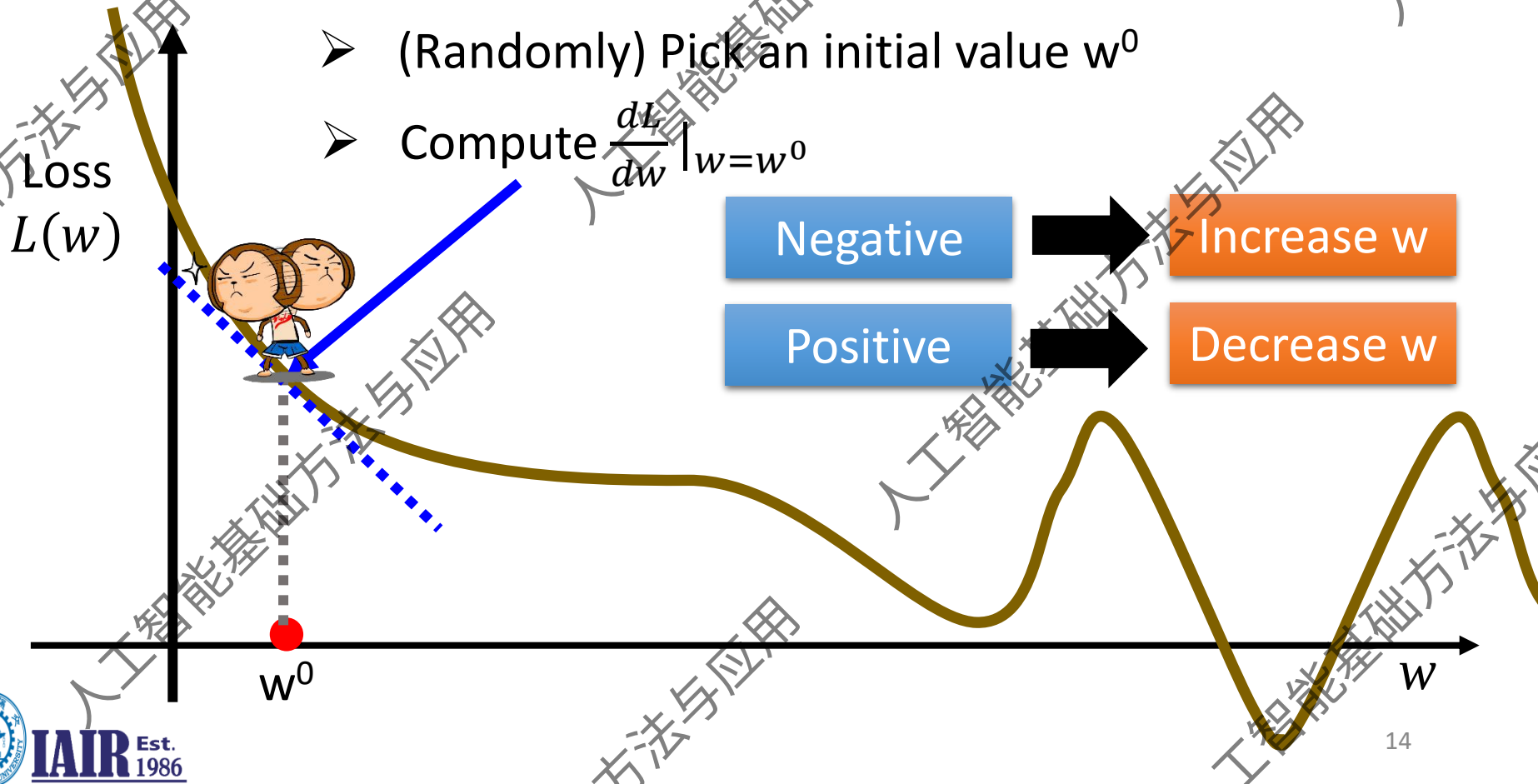


人工智能基础方法与应用

梯度下降：先判断往哪边走

- Consider loss function $L(w)$ with one parameter w:

$$w^* = arg\min_w L(w)$$

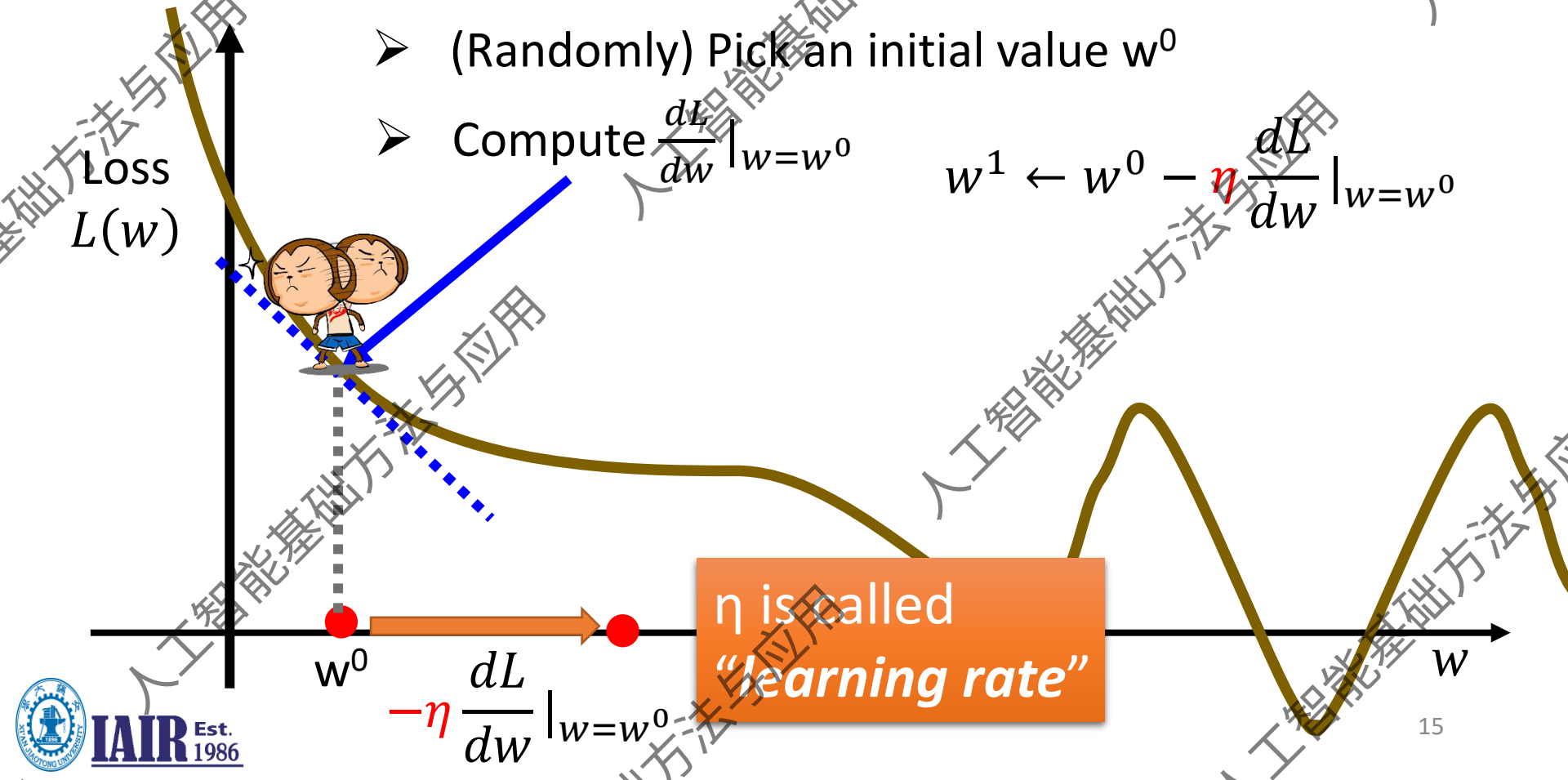


人工智能基础方法与应用

学习率决定每一步的大小

- Consider loss function $L(w)$ with one parameter w:

$$w^* = arg\min_w L(w)$$

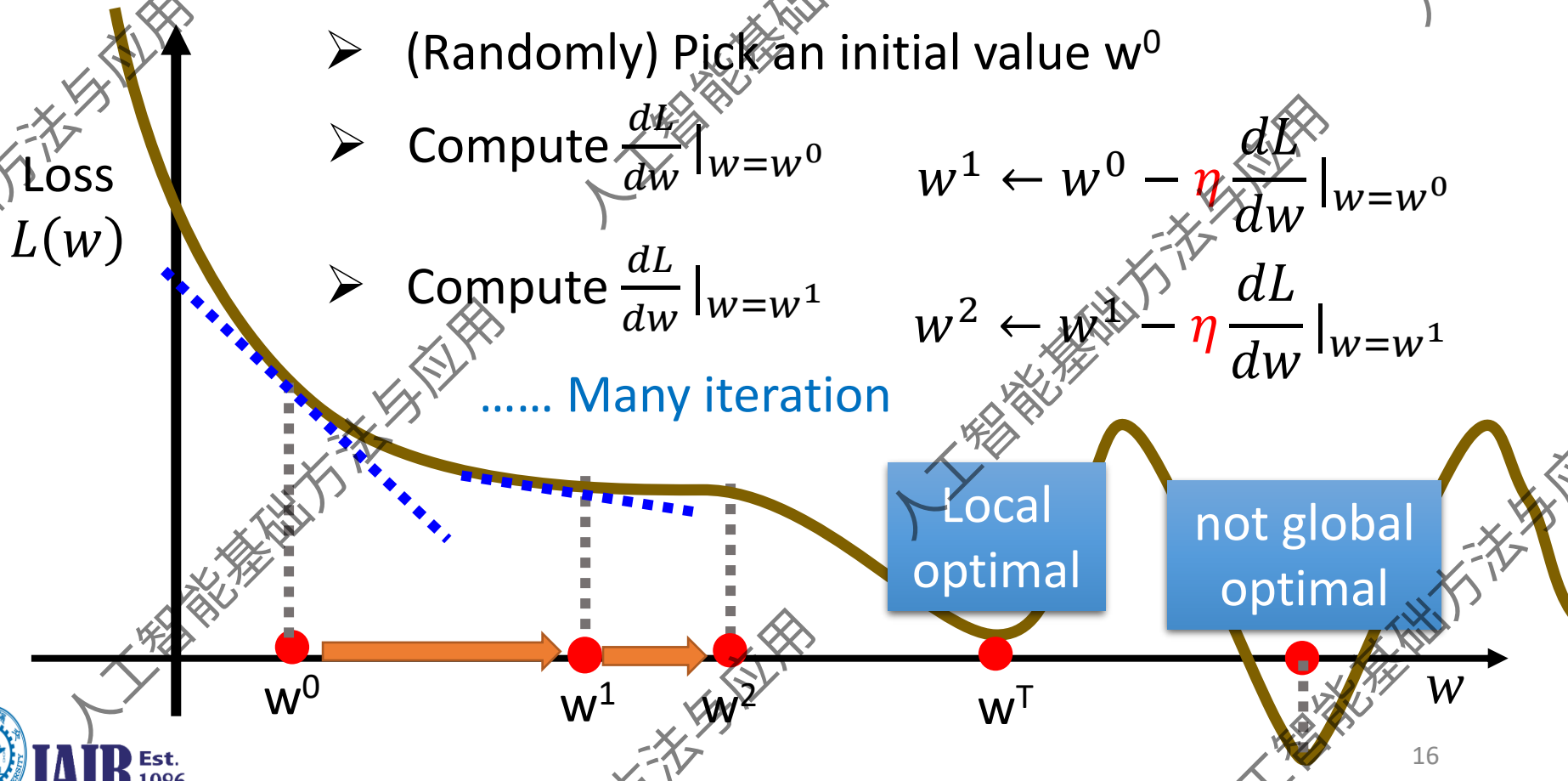


人工智能基础方法与应用

反复更新直到损失不再明显下降

- Consider loss function $L(w)$ with one parameter w:

$$w^* = arg\min_w L(w)$$



Step 3

$$\nabla L = \begin{bmatrix} \frac{\partial L}{\partial w} \\ \frac{\partial L}{\partial b} \end{bmatrix}$$ gradient

- How about two parameters?

$$w^*, b^* = arg \min_{w,b} L(w,b)$$

- (Randomly) Pick an initial value w^0, b^0
- Compute $\frac{\partial L}{\partial w}\Big|_{w=w^0,b=b^0}$, $\frac{\partial L}{\partial b}\Big|_{w=w^0,b=b^0}$

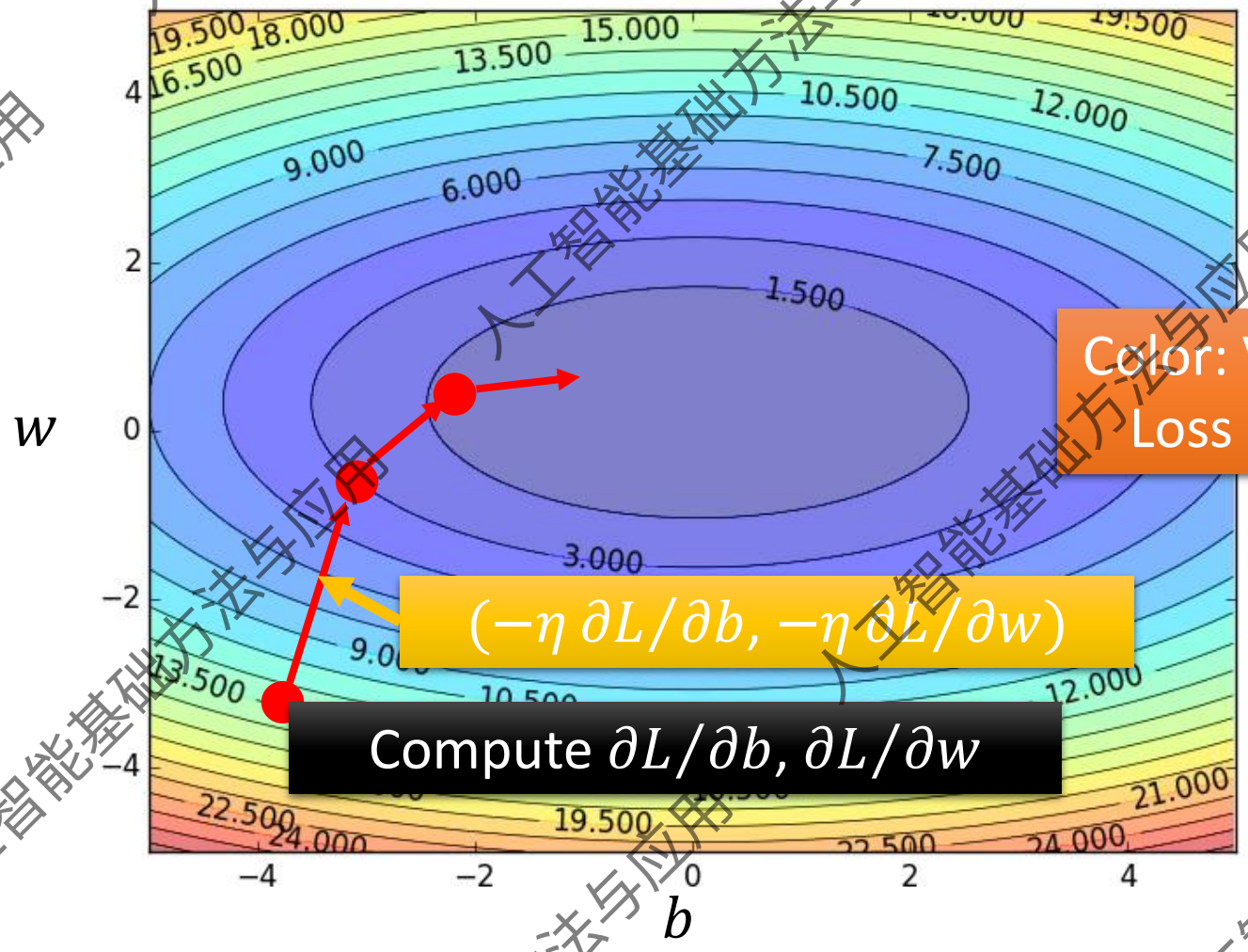
$$w^1 \leftarrow w^0 - \eta \frac{\partial L}{\partial w}\Big|_{w=w^0,b=b^0} \qquad b^1 \leftarrow b^0 - \eta \frac{\partial L}{\partial b}\Big|_{w=w^0,b=b^0}$$

- Compute $\frac{\partial L}{\partial w}\Big|_{w=w^1,b=b^1}$, $\frac{\partial L}{\partial b}\Big|_{w=w^1,b=b^1}$

$$w^2 \leftarrow w^1 - \eta \frac{\partial L}{\partial w}\Big|_{w=w^1,b=b^1} \qquad b^2 \leftarrow b^1 - \eta \frac{\partial L}{\partial b}\Big|_{w=w^1,b=b^1}$$

人工智能基础方法与应用

在二维损失地形下降

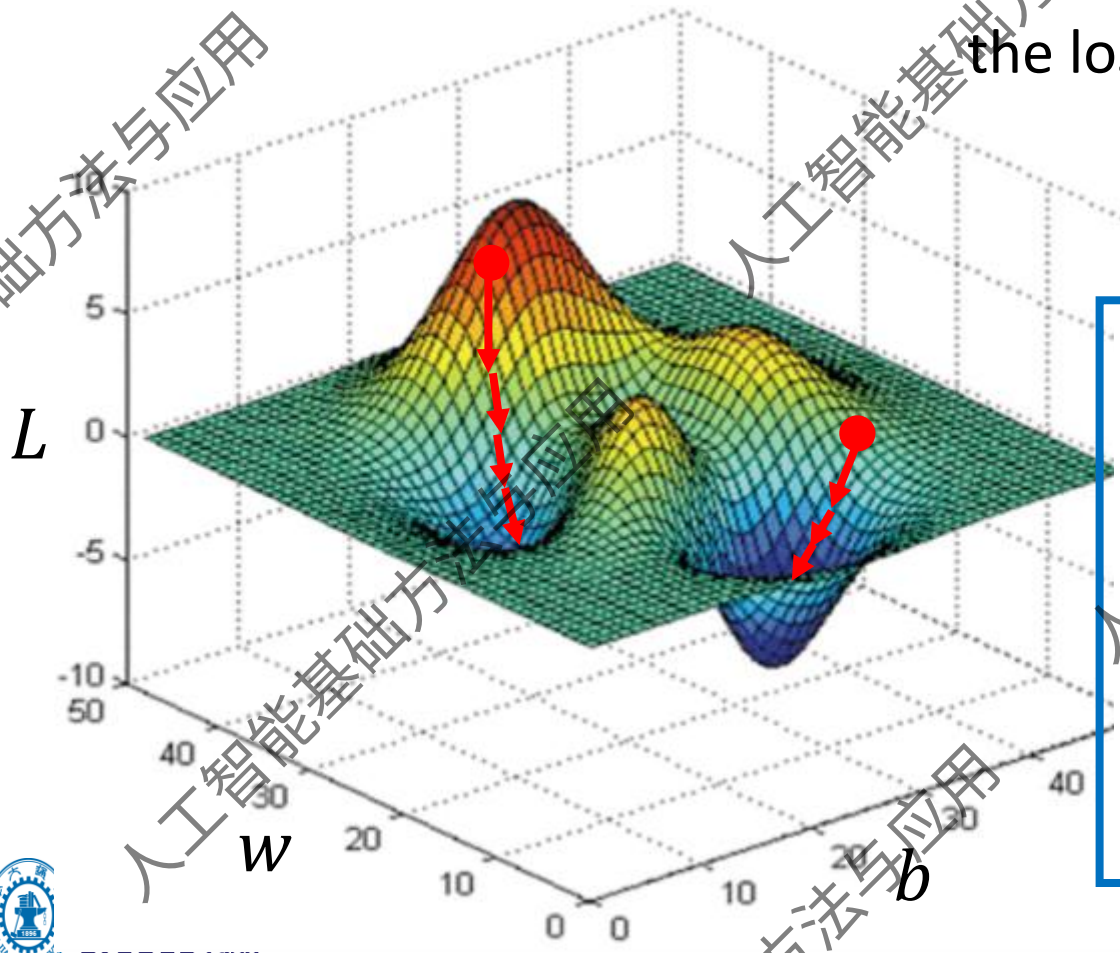

19.500
18.000
16.500
15.000
13.500
10.500
12.000
9.000
7.500
6.000
1.500
3.000
22.500
24.000
19.500
21.000
4
2
0
−2
−4
−4
−2
0
2
4
w
b
Color: Value of Loss L(w,b)
$(-\eta\,\partial L/\partial b, -\eta\,\partial L/\partial w)$
Compute $\partial L/\partial b, \partial L/\partial w$

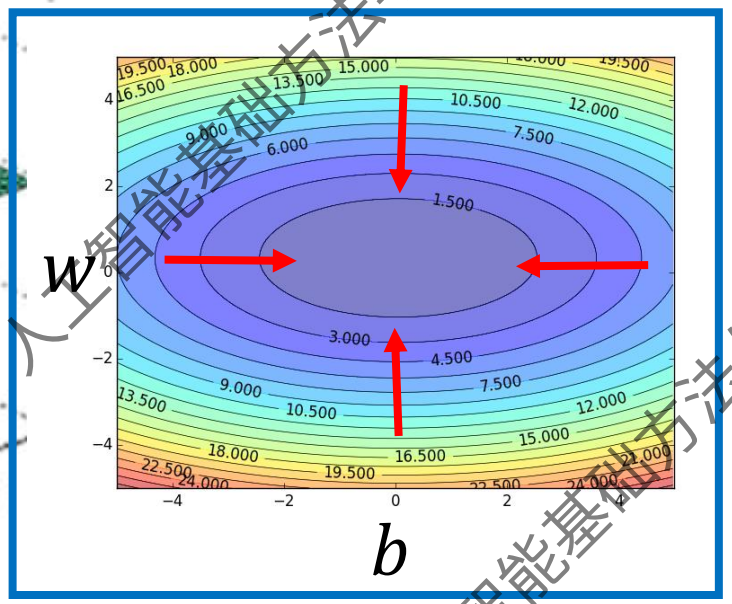
Step 3: Gradient Descent

- Worry?

Don't worry. In linear regression, the loss function L is convex.

No local optimal





先看训练数据上的结果

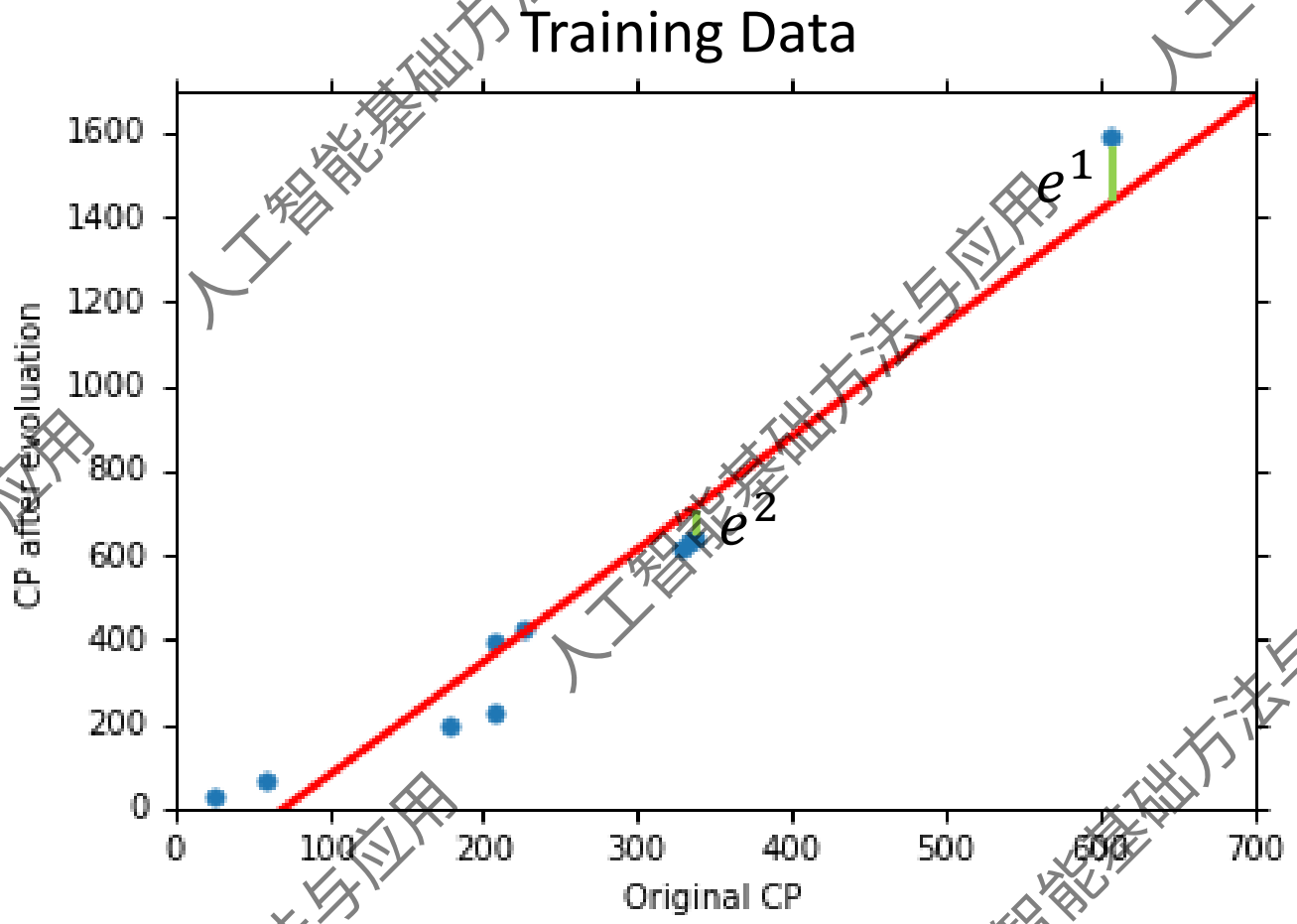
$y = b + w \cdot x_{cp}$

b = -188.4

w = 2.7

Average Error on Training Data

$$= \sum_{n=1}^{10} e^n = 31.9$$



人工智能基础方法与应用

泛化性

$y = b + w \cdot x_{cp}$

b = -188.4

w = 2.7

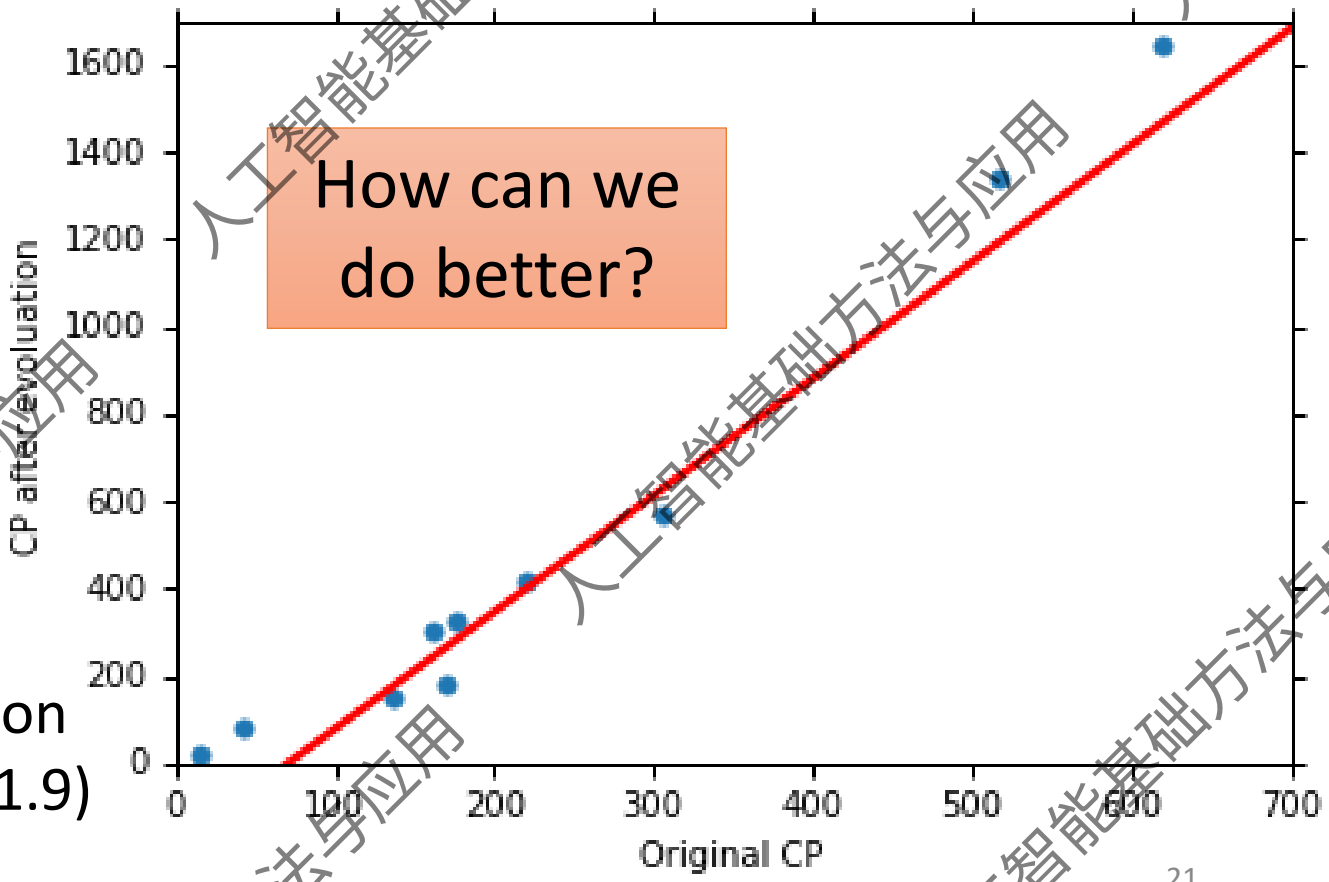
Average Error on Testing Data

$$= \sum_{n=1}^{10} e^n = 35.0$$

> Average Error on Training Data (31.9)

What we really care about is the error on new data (testing data)

Another 10 pokemons as testing data



人工智能基础方法与应用

Selecting another Model

$$y = b + w_1 \cdot x_{cp} + w_2 \cdot (x_{cp})^2$$

Best Function

b = -10.3

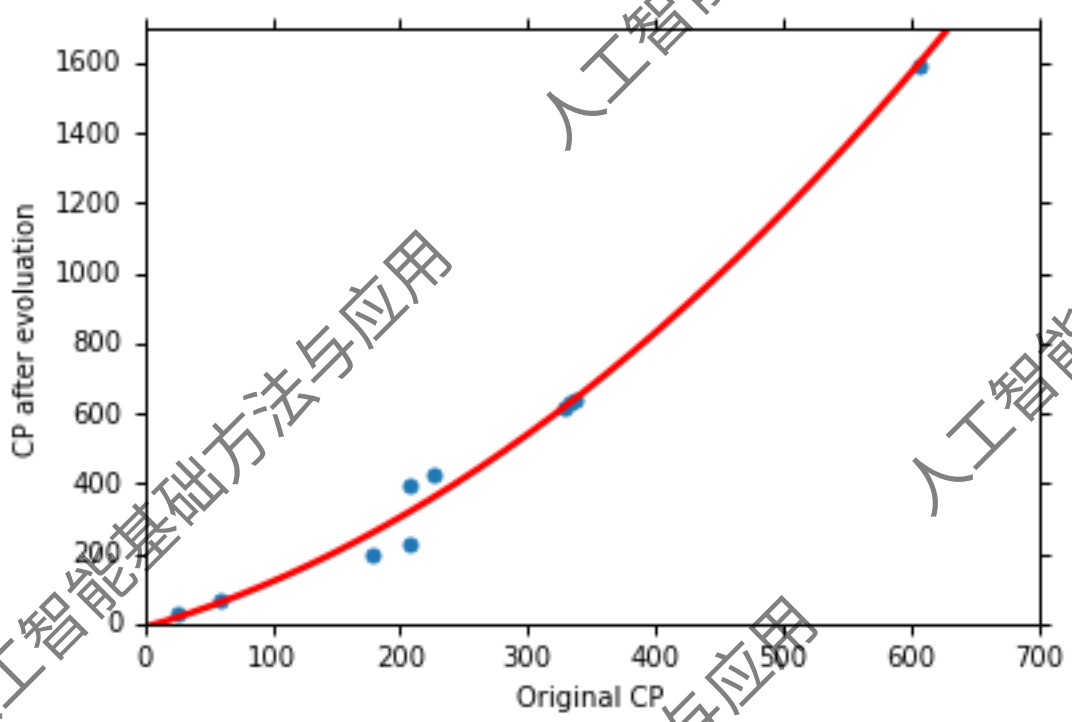
$w_1 = 1.0$, $w_2 = 2.7 \times 10^{-3}$

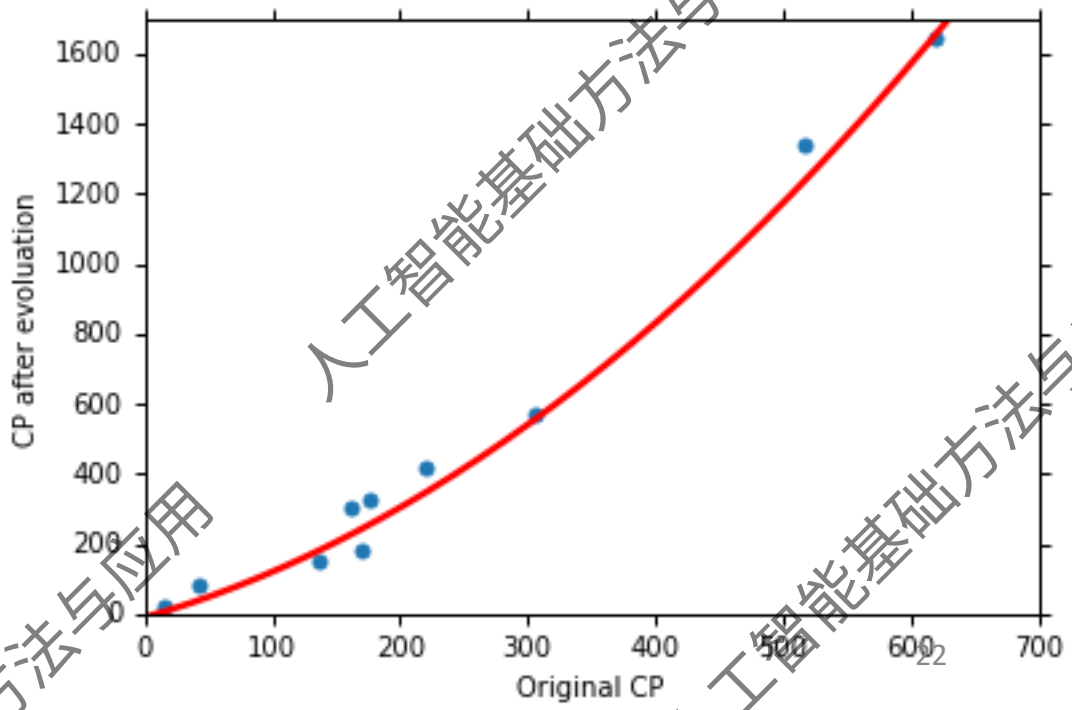
Average Error = 15.4

Testing:

Average Error = 18.4

Better! Could it be even better?





人工智能基础方法与应用

Selecting another Model

$$y = b + w_1 \cdot x_{cp} + w_2 \cdot (x_{cp})^2 + w_3 \cdot (x_{cp})^3$$

Best Function

$b = 6.4$, $w_1 = 0.66$

$w_2 = 4.3 \times 10^{-3}$

$w_3 = -1.8 \times 10^{-6}$

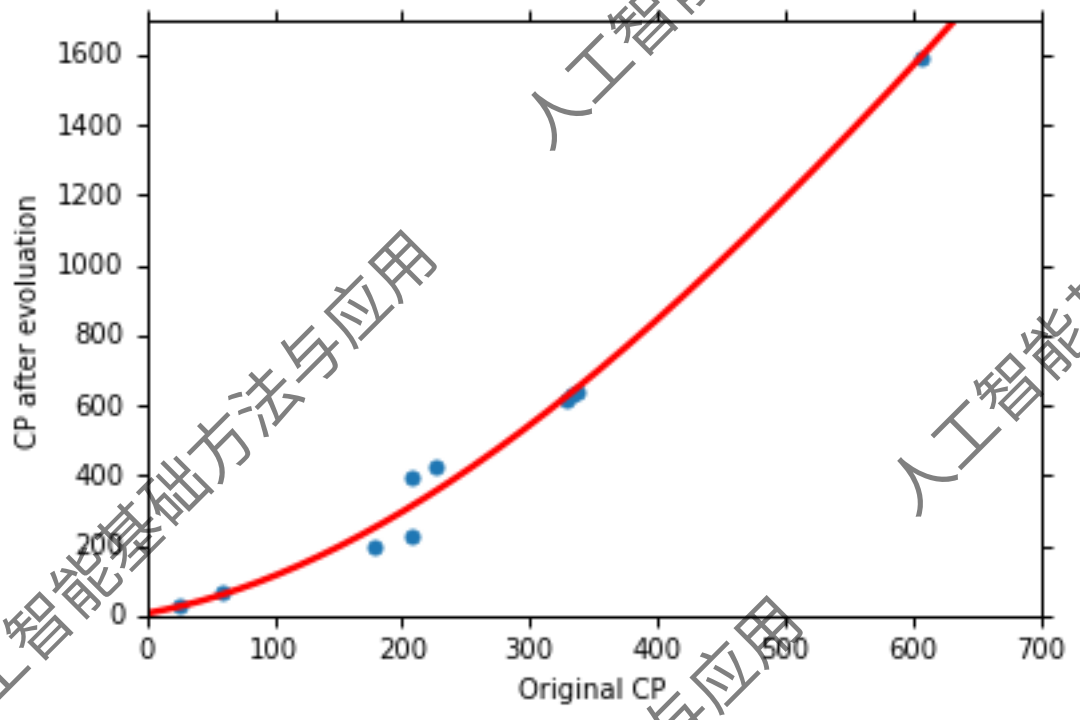
Average Error = 15.3

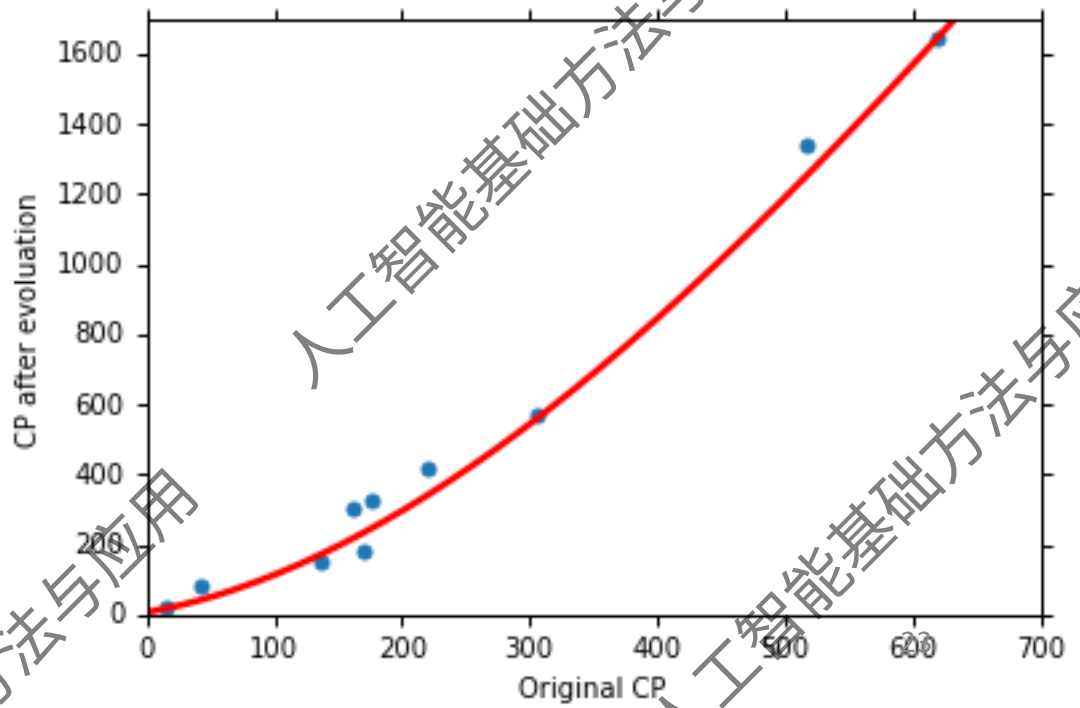
Testing:

Average Error = 18.1

Slightly better.

How about more complex model?





人工智能基础方法与应用

Selecting another Model

$$y = b + w_1 \cdot x_{cp} + w_2 \cdot (x_{cp})^2 + w_3 \cdot (x_{cp})^3 + w_4 \cdot (x_{cp})^4$$

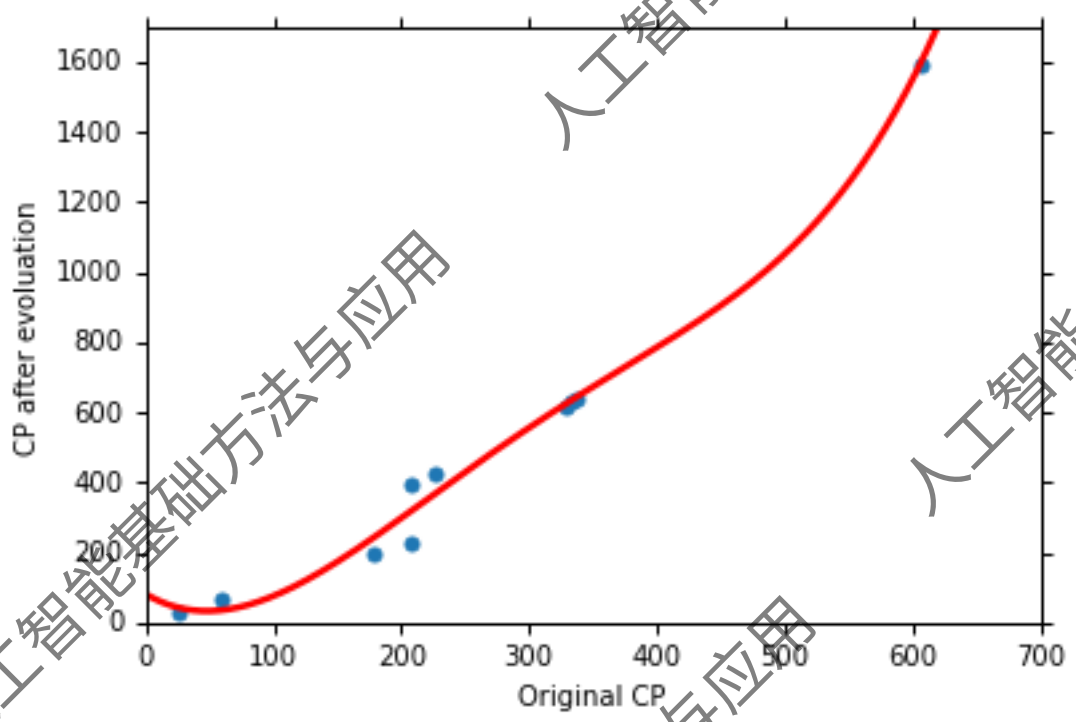
Best Function

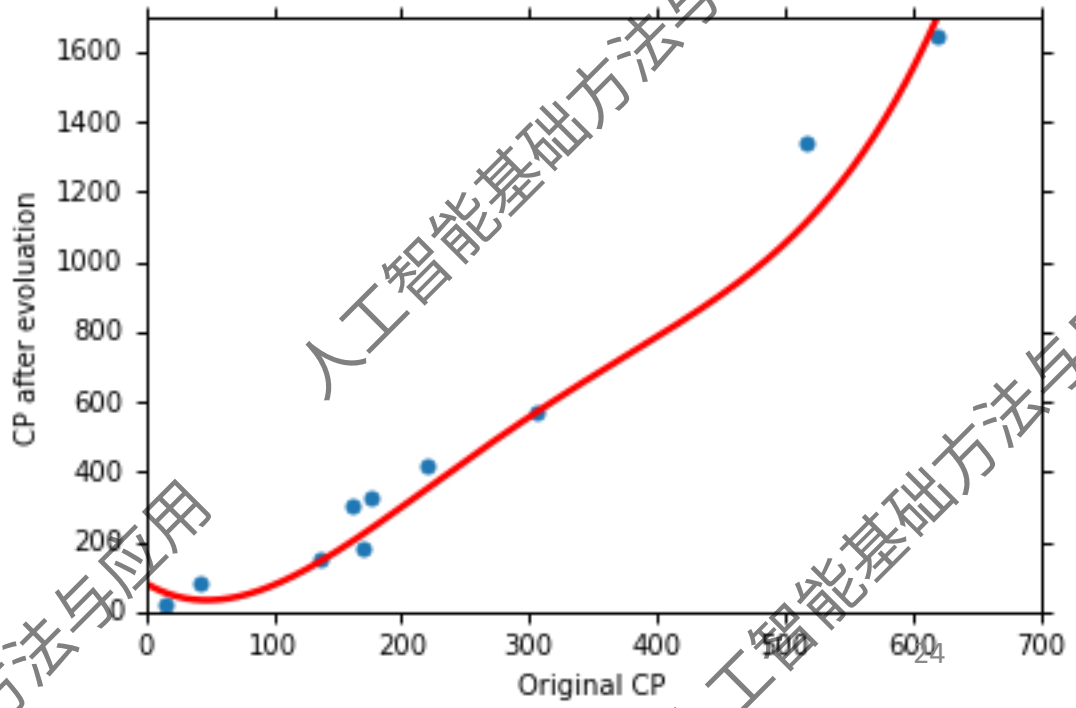
Average Error = 14.9

Testing:

Average Error = 28.8

The results become worse ...

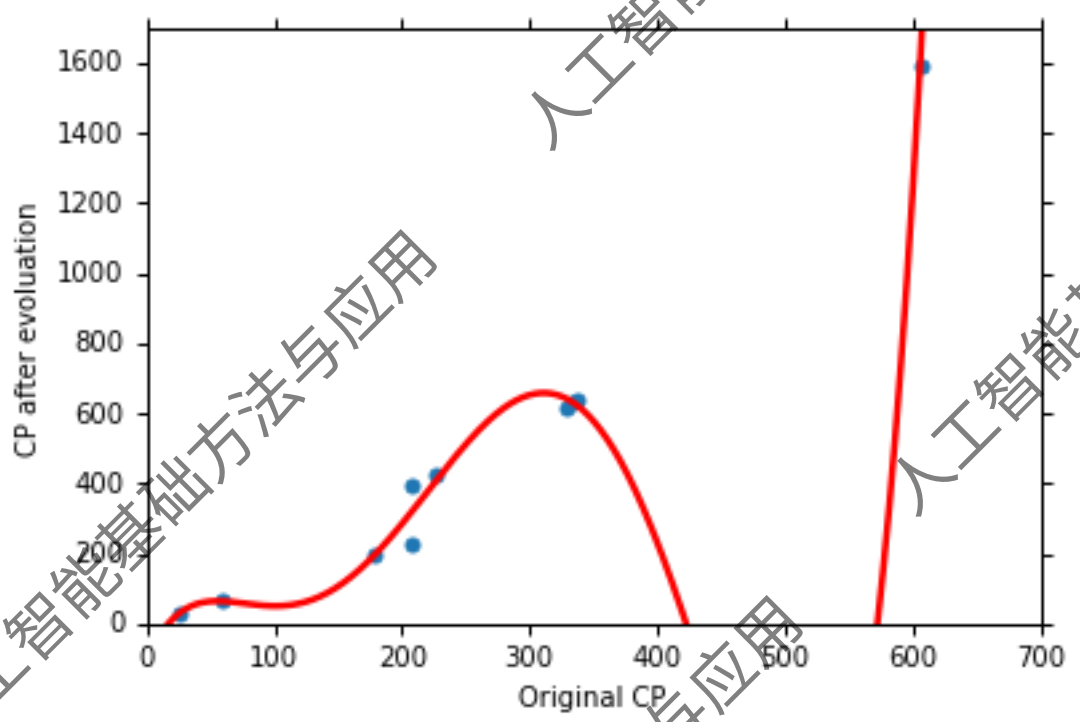




人工智能基础方法与应用

Selecting another Model

$$y = b + w_1 \cdot x_{cp} + w_2 \cdot (x_{cp})^2 + w_3 \cdot (x_{cp})^3 + w_4 \cdot (x_{cp})^4 + w_5 \cdot (x_{cp})^5$$



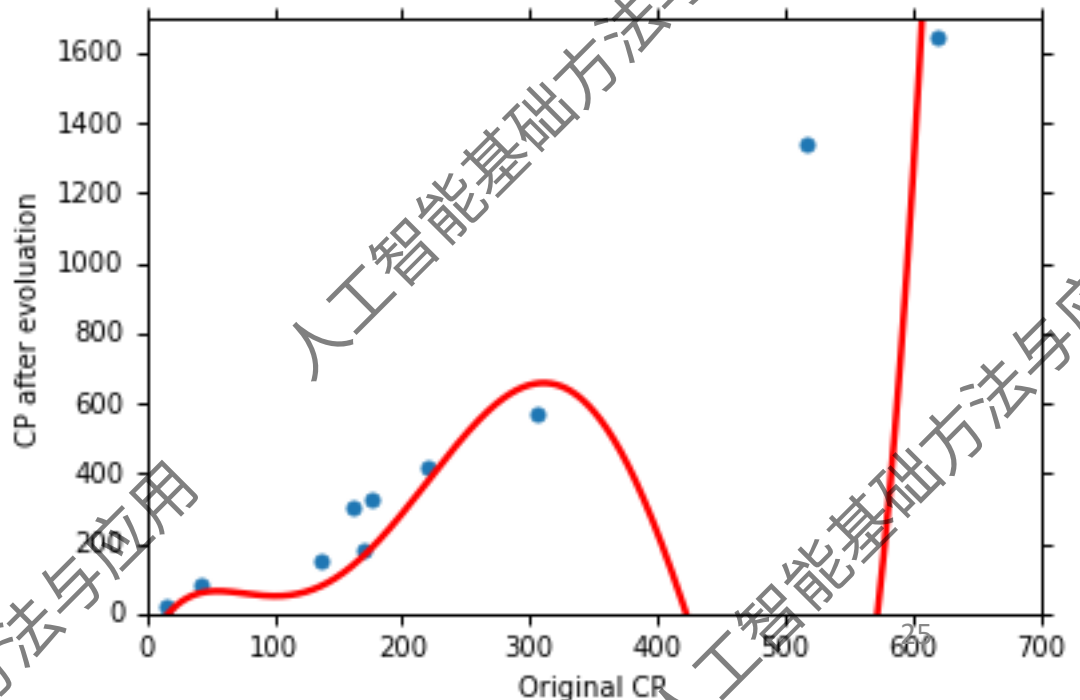
Best Function

Average Error = 12.8

Testing:

Average Error = 232.1

The results are so bad.



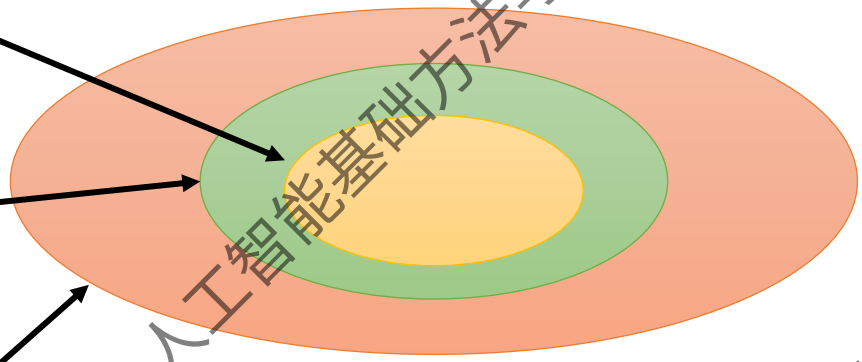
人工智能基础方法与应用

Model Selection

Training Data



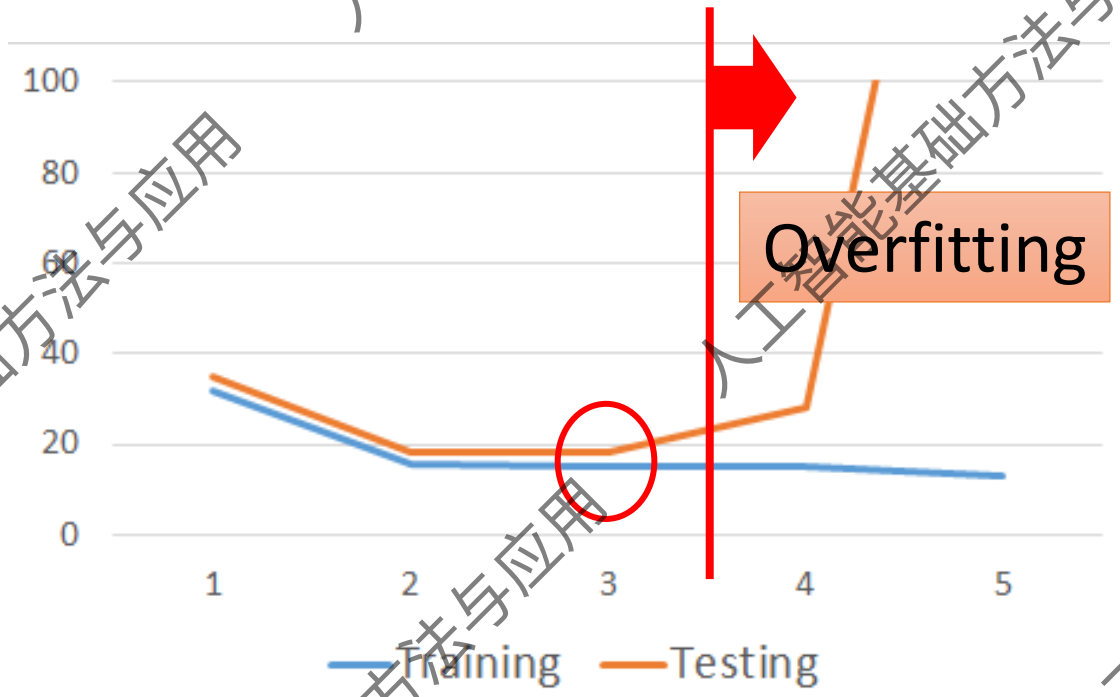
1. $y = b + w \cdot x_{cp}$
2. $y = b + w_1 \cdot x_{cp} + w_2 \cdot (x_{cp})^2$
3. $y = b + w_1 \cdot x_{cp} + w_2 \cdot (x_{cp})^2 + w_3 \cdot (x_{cp})^3$
4. $y = b + w_1 \cdot x_{cp} + w_2 \cdot (x_{cp})^2 + w_3 \cdot (x_{cp})^3 + w_4 \cdot (x_{cp})^4$
5. $y = b + w_1 \cdot x_{cp} + w_2 \cdot (x_{cp})^2 + w_3 \cdot (x_{cp})^3 + w_4 \cdot (x_{cp})^4 + w_5 \cdot (x_{cp})^5$



A more complex model yields lower error on training data.

If we can truly find the best function

Model Selection



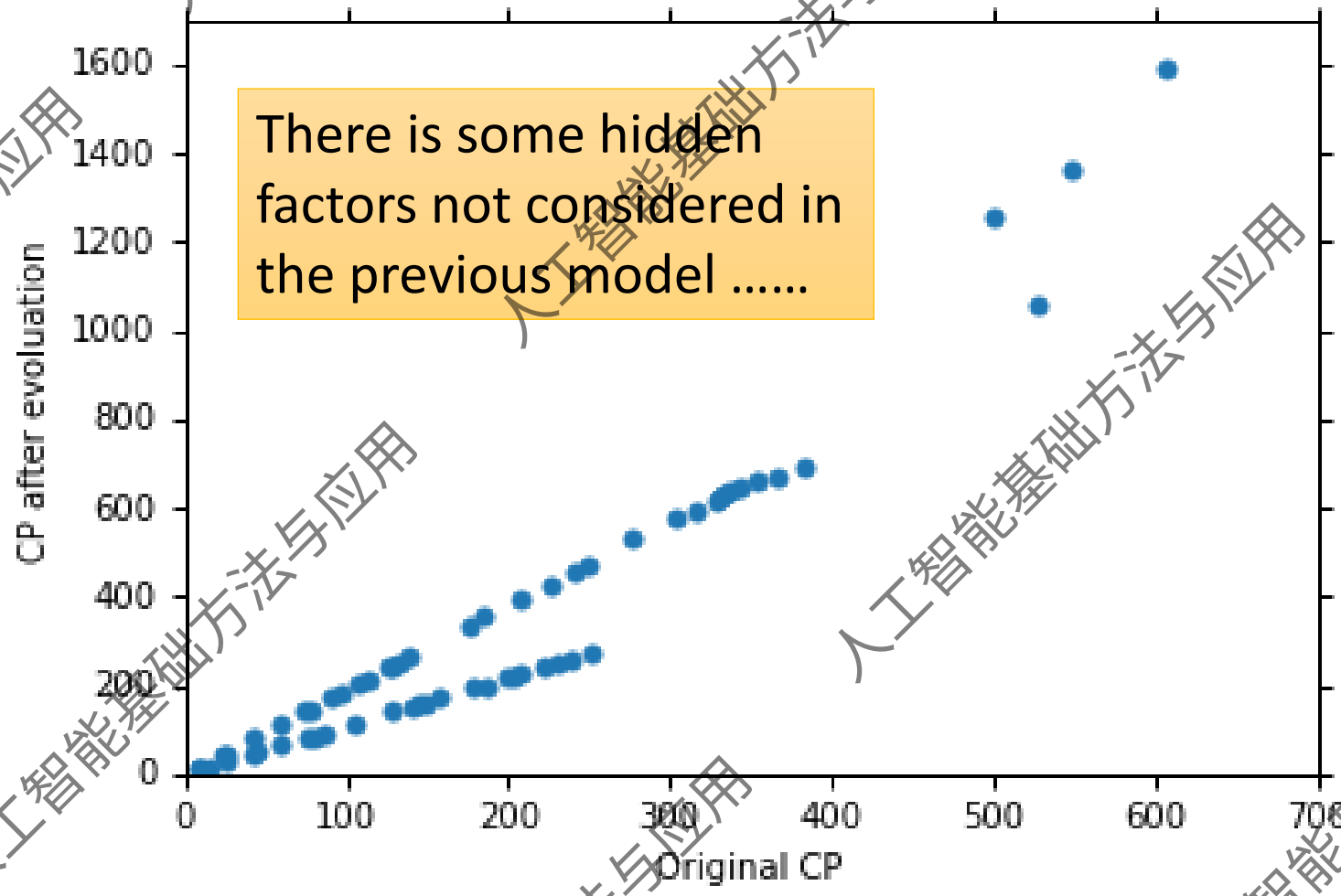
	Training	Testing
1	31.9	35.0
2	15.4	18.4
3	15.3	18.1
4	14.9	28.2
5	12.8	232.1

A more complex model does not always lead to better performance on ***testing data***.

This is ***Overfitting***. → Select suitable model

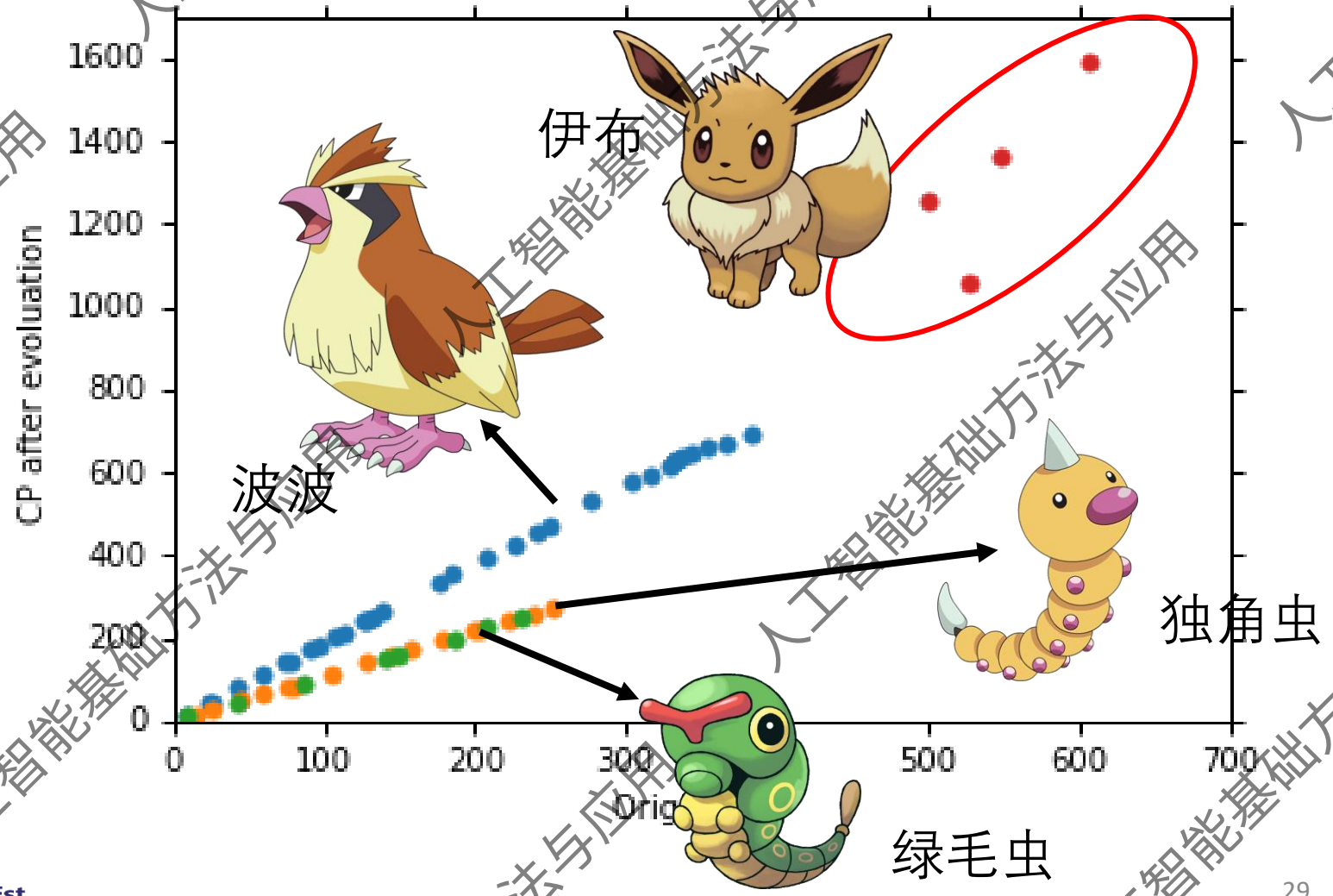
人工智能基础方法与应用

Let’s collect more data



人工智能基础方法与应用

What are the hidden factors?



人工智能基础方法与应用

Back to step 1: Redesign the Model

$$y = b + \sum w_i x_i$$

Linear model?

x_s = species of x

x

If x_s = 波波: $y = b_1 + w_1 \cdot x_{cp}$

If x_s = 独角虫: $y = b_2 + w_2 \cdot x_{cp}$

If x_s = 绿毛虫: $y = b_3 + w_3 \cdot x_{cp}$

If x_s = 伊布: $y = b_4 + w_4 \cdot x_{cp}$

y

人工智能基础方法与应用

Back to step 1: Redesign the Model

$$y = b + \sum w_i x_i$$

Linear model?

$$y = b_1 \cdot 1 + w_1 \cdot 1 + b_2 \cdot 0 + w_2 \cdot 0 + b_3 \cdot 0 + w_3 \cdot 0 + b_4 \cdot 0 + w_4 \cdot 0$$

$$\delta(x_s = \text{波波})$$

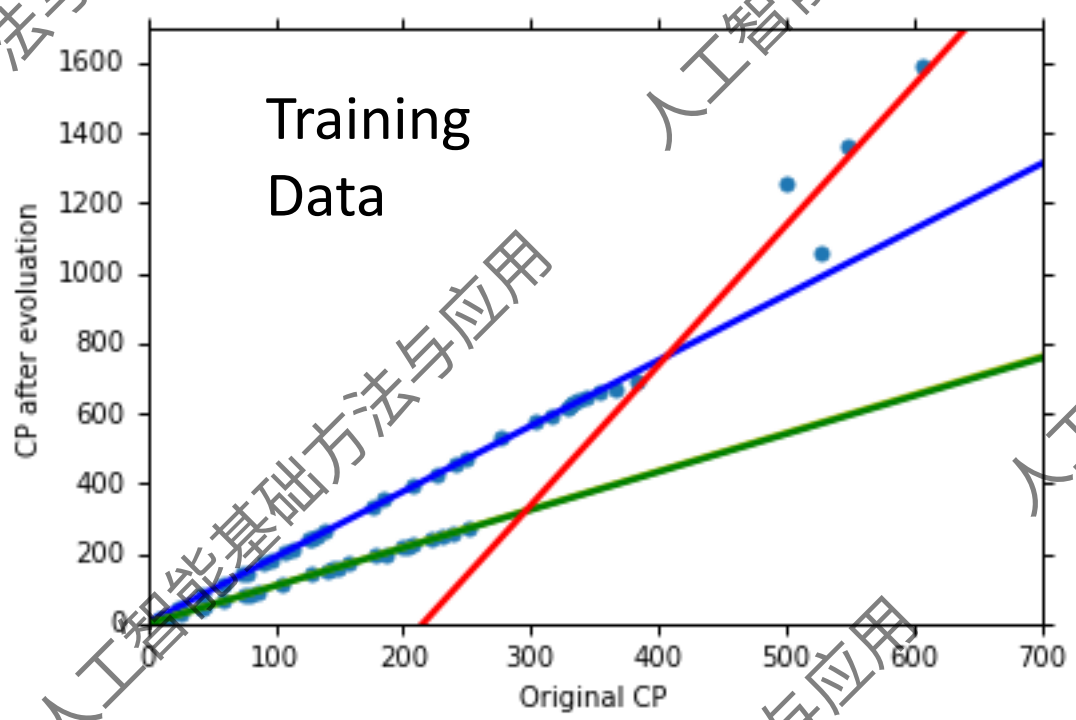
$$\begin{cases} =1 & \text{If } x_s = \text{波波} \\ =0 & \text{otherwise} \end{cases}$$

If $x_s = \text{波波}$

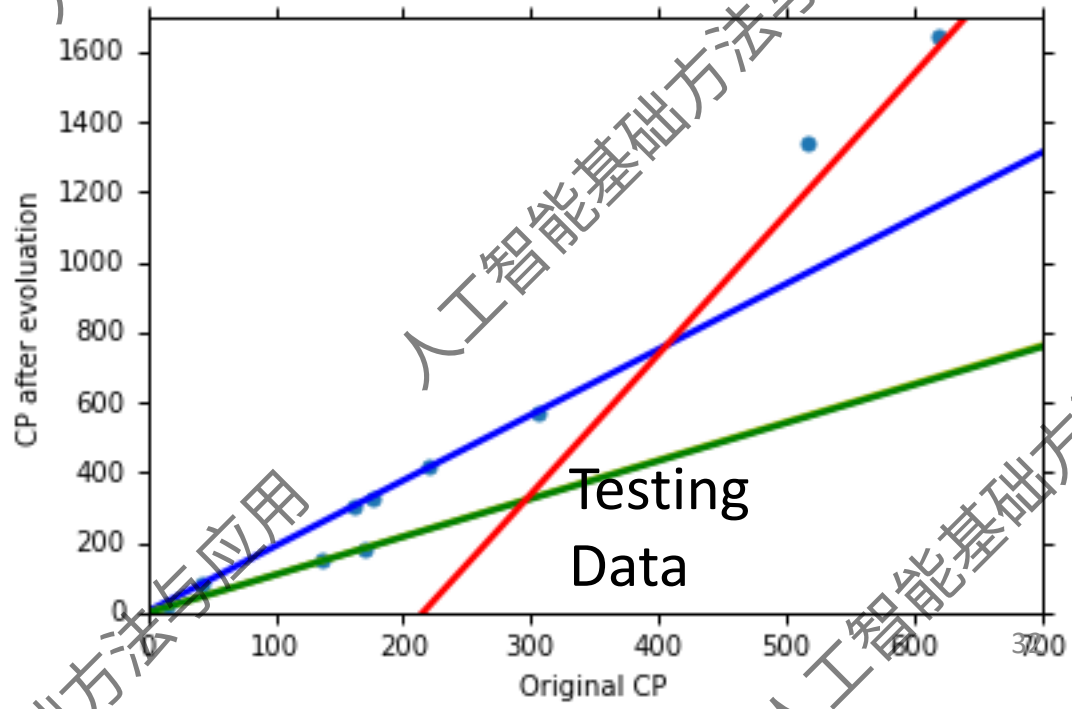
$$y = b_1 + w_1 \cdot x_{cp}$$

人工智能基础方法与应用

Average error
= 3.8

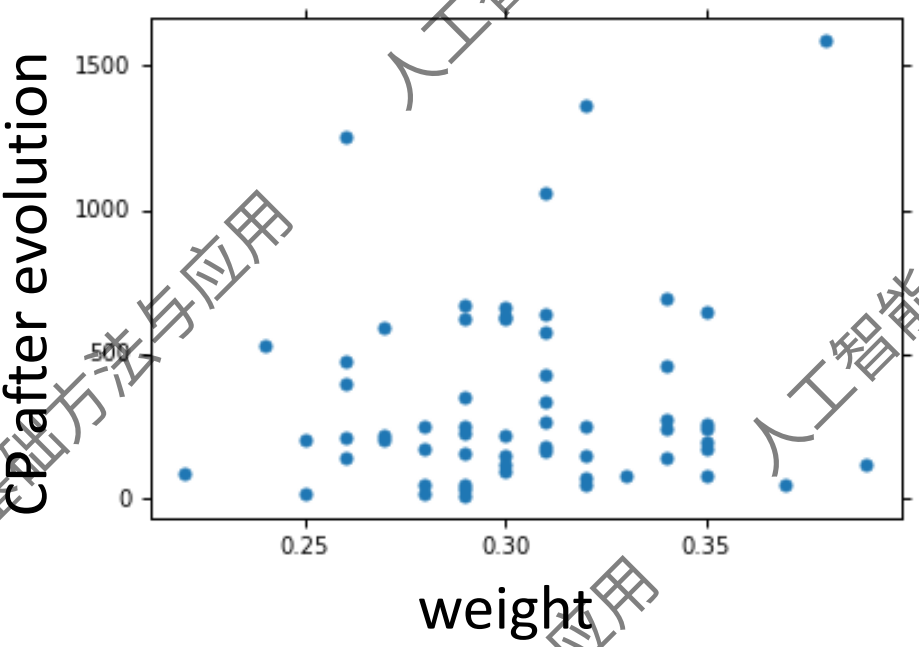


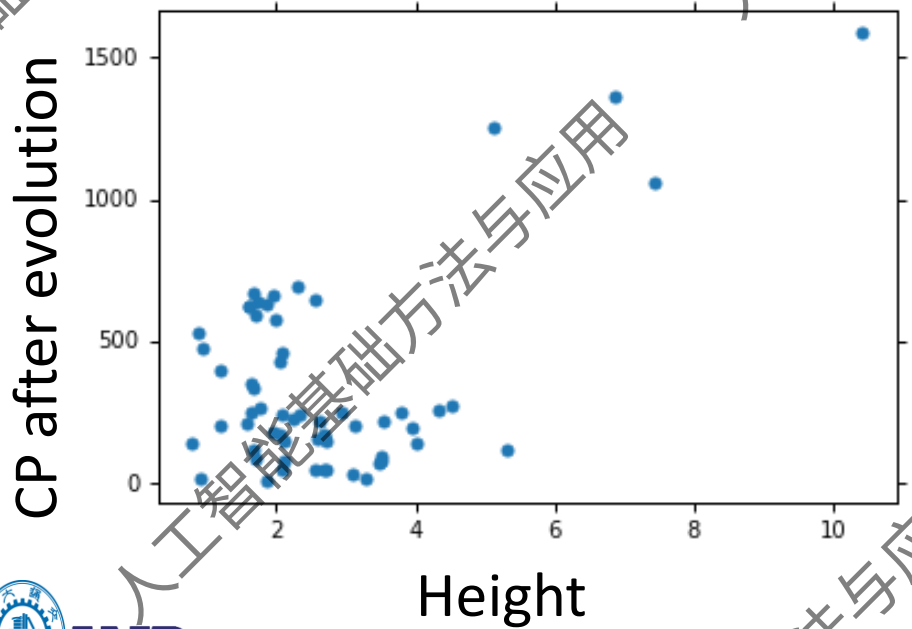
Average error
= 14.3

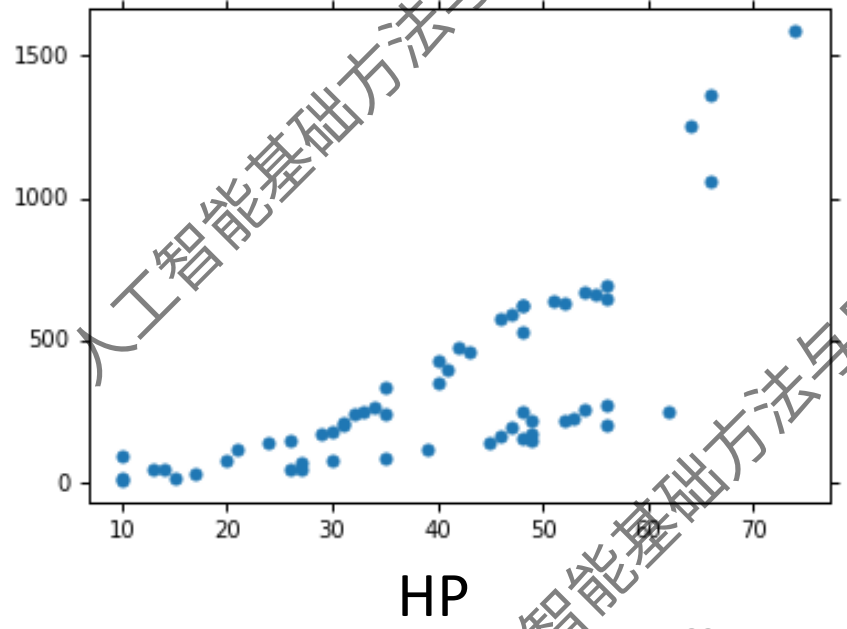


人工智能基础方法与应用

Are there any other hidden factors?

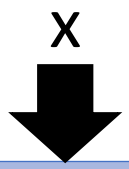






人工智能基础方法与应用

Back to step 1: Redesign the Model Again



If $x_s =$ 波波: $y' = b_1 + w_1 \cdot x_{cp} + w_5 \cdot (x_{cp})^2$

If $x_s =$ 独角虫: $y' = b_2 + w_2 \cdot x_{cp} + w_6 \cdot (x_{cp})^2$

If $x_s =$ 绿毛虫: $y' = b_3 + w_4 \cdot x_{cp} + w_7 \cdot (x_{cp})^2$

If $x_s =$ 伊布: $y' = b_4 + w_4 \cdot x_{cp} + w_8 \cdot (x_{cp})^2$

$$y = y' + w_9 \cdot x_{hp} + w_{10} \cdot (x_{hp})^2 + w_{11} \cdot x_h + w_{12} \cdot (x_h)^2 + w_{13} \cdot x_w + w_{14} \cdot (x_w)^2$$



Training Error = 1.9

Testing Error = 102.3

Overfitting!

人工智能基础方法与应用

Back to step 2: Regularization

$$y = b + \sum w_i x_i$$

对过大的非偏置权重收取额外代价。使得 w_i 越小越好

$$L = \sum_n \left(\hat{y}^n - \left(b + \sum w_i x_i \right) \right)^2 \quad \boxed{+\lambda \sum (w_i)^2}$$

- Why smooth functions are preferred?

$$y = b + \sum w_i x_i$$

$+w_i \Delta x_i$ $+\Delta x_i$ 噪声

- If some noises corrupt input x_i when testing

A smoother function has less influence.

人工智能基础方法与应用

Regularization

λ	Training	Testing
0	1.9	102.3
1	2.3	68.7
10	3.5	25.7
100	4.1	11.1
1000	5.6	12.8
10000	6.3	18.7
100000	8.5	26.8

smoother



How smooth?

Select λ obtaining the best model

- Training error: largerλ, considering the training error less
- We prefer smooth function, but don't be too smooth.

Conclusion

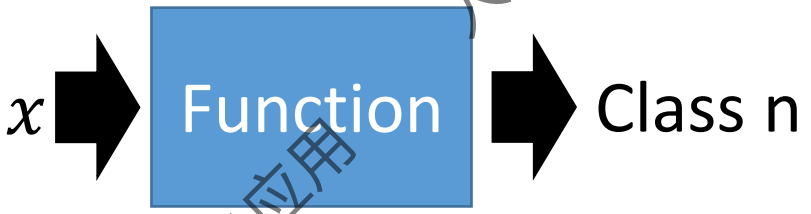
- 宝可梦进化预测
- 梯度下降
- 过拟合与正则化
- 模型最终在测试集上的平均误差约为 11.1
  - 如果换一批新的测试数据，结果还会一样吗？
  - 当前误差会不会是低估或高估了模型真实表现？

人工智能基础方法与应用

二、分类：概率生成模型

人工智能基础方法与应用

Classification



- 垃圾邮件识别
  - Input: 邮件内容、标题、发件人、关键词等
  - Output: 垃圾邮件 / 正常邮件
- 医疗诊断
  - Input: 当前症状、年龄、性别、既往病史等
  - Output: 疾病类型
- 手写字符识别

Input:  output: **金**

- 人脸识别
  - Input: 人脸图像, output: 人物身份

人工智能基础方法与应用

应用示例：宝可梦属性预测



$f($  $) =$  $f($  $) =$  $f($  $) =$ 

应用示例



- **总和（Total）**：后面各项能力值的总和，可作为衡量一只宝可梦整体强弱的大致指标 320
- **生命值（HP）**：即生命点数，表示宝可梦在倒下前能够承受多少伤害 35
- **攻击（Attack）**：普通物理攻击的基础能力值，例如抓、拳击等 55
- **防御（Defense）**：抵抗普通物理攻击伤害的基础能力值 40
- **特攻（SP Atk, Special Attack）**：特殊攻击的基础能力值，例如“十万伏特”“喷射火焰” 50
- **特防（SP Def, Special Defense）**：抵抗特殊攻击伤害的基础能力值 50
- **速度（Speed）**：通常决定一回合中哪只宝可梦先行动 90

Can we predict the “type” of pokemon based on the information?

How to do Classification

- Training data for Classification

$(x^1, \hat{y}^1)$ $(x^2, \hat{y}^2)$ …… $(x^N, \hat{y}^N)$

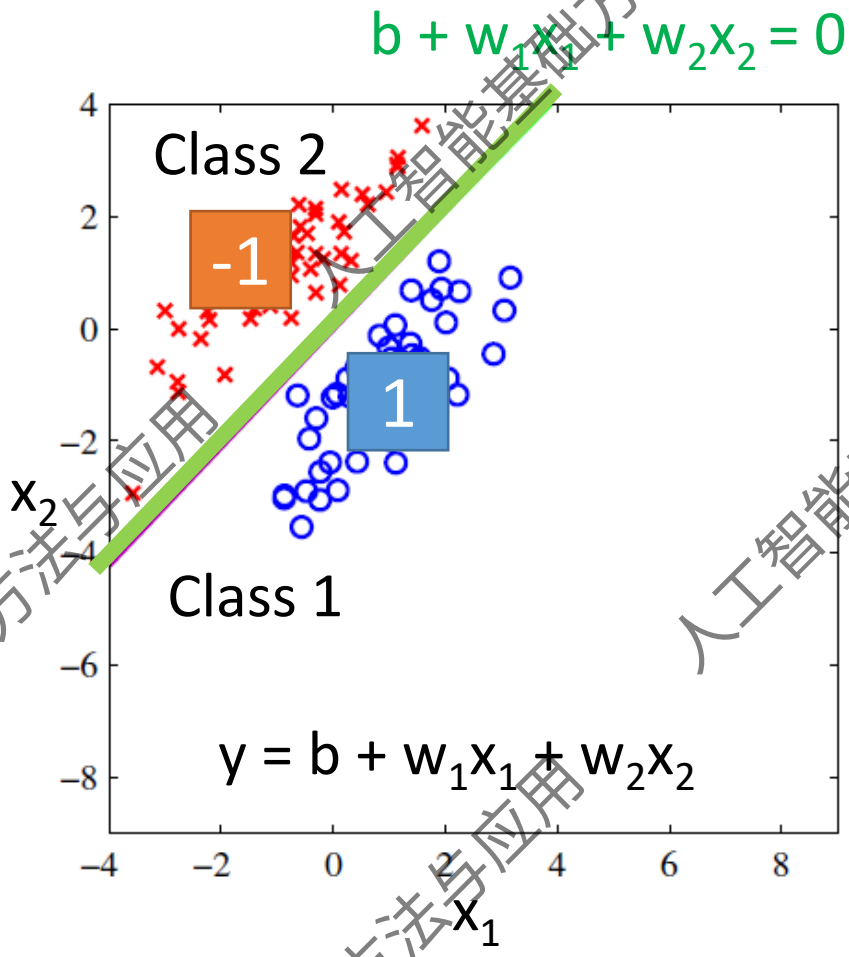


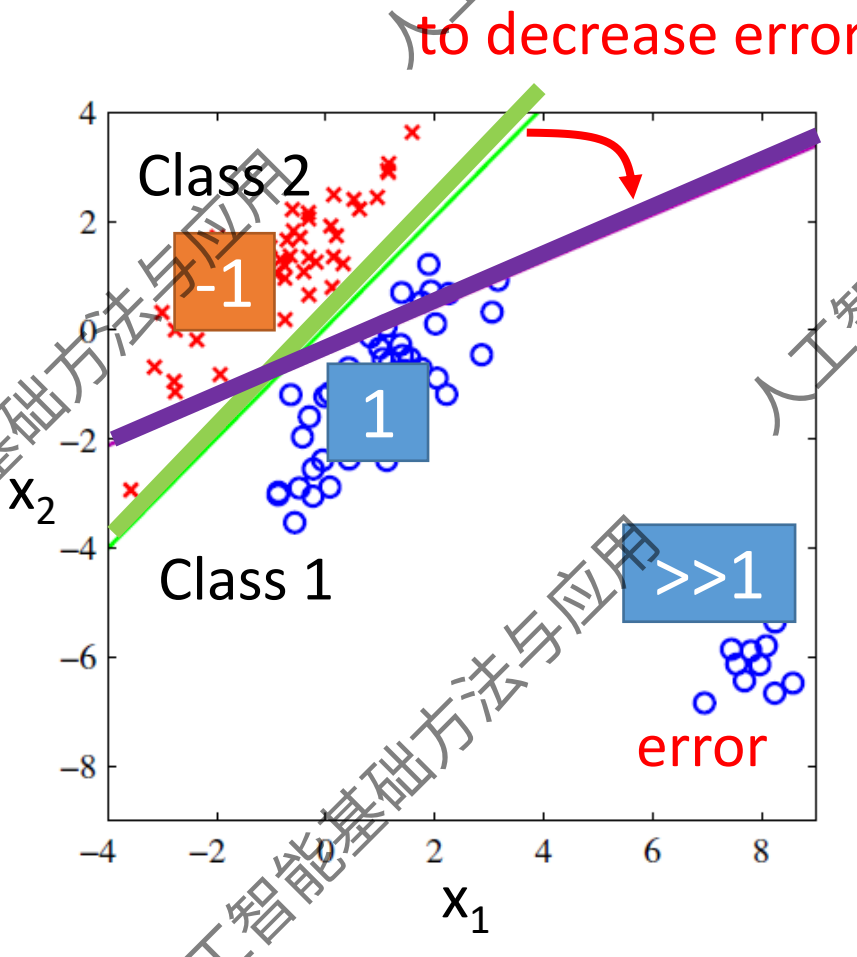
Classification as Regression?

Binary classification as example

Training: Class 1 means the target is 1; Class 2 means the target is -1

Testing: closer to 1 → class 1; closer to -1 → class 2





Penalize to the examples that are "too correct" …

- Multiple class: Class 1 means the target is 1; Class 2 means the target is 2; Class 3 means the target is 3 …… problematic

Ideal Alternatives

- Function (Model): $f(x)$

x ➡

$g(x) > 0$	Output = class 1
$else$	Output = class 2

- Loss function:

$$L(f) = \sum_n \delta(f(x^n) \neq \hat{y}^n)$$

The number of times f get incorrect results on training data.

- Find the best function:
  - Example: Perceptron, SVM

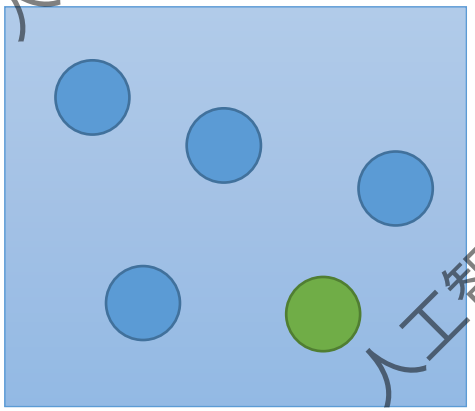
Not Today

人工智能基础方法与应用

从抽样问题理解贝叶斯判断

Box 1

$P(B_1) = 2/3$

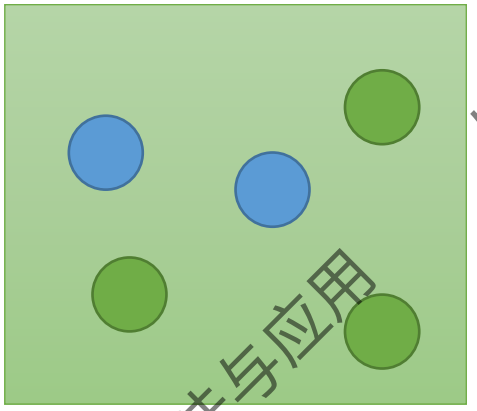


$P(\text{Blue}|B_1) = 4/5$

$P(\text{Green}|B_1) = 1/5$

Box 2

$P(B_2) = 1/3$



$P(\text{Blue}|B_2) = 2/5$

$P(\text{Green}|B_2) = 3/5$

from one of the boxes

Where does it come from?

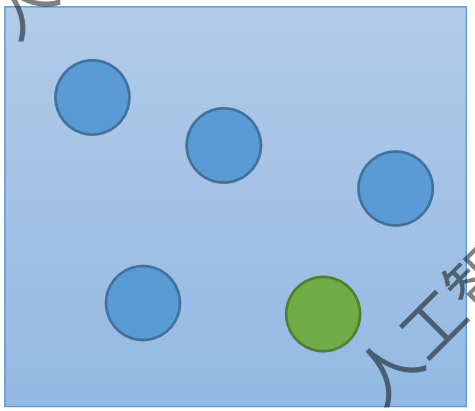
$$P(B_1 \mid \text{Blue}) = \frac{P(\text{Blue}|B_1)P(B_1)}{P(\text{Blue}|B_1)P(B_1) + P(\text{Blue}|B_2)P(B_2)}$$

把“盒子”换成“类别”

Estimating the Probabilities
From training data

Class 1

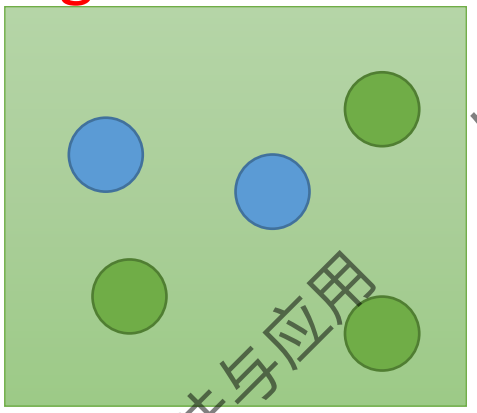
$P(C_1)$



$P(x|C_1)$

Class 2

$P(C_2)$



$P(x|C_2)$

Given an x, which class does it belong to

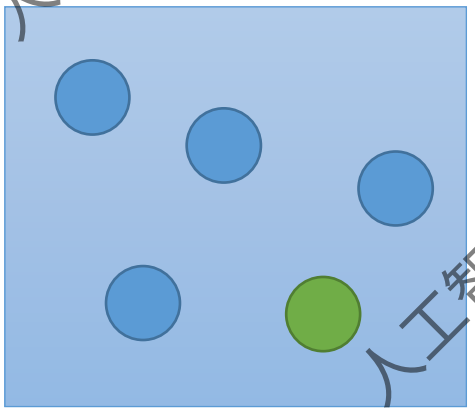
$$P(C_1|x) = \frac{P(x|C_1)P(C_1)}{P(x|C_1)P(C_1) + P(x|C_2)P(C_2)}$$

Generative Model $P(x) = P(x|C_1)P(C_1) + P(x|C_2)P(C_2)$

先验来自类别出现频率

Class 1

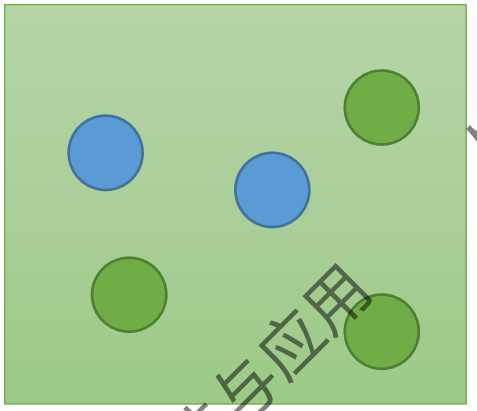
$P(C_1)$



Water

Class 2

$P(C_2)$



Normal

Water and Normal type with 140 for training, rest for testing

Training: 79 Water, 61 Normal

$P(C_1) = 79 / (79 + 61) = 0.56$

$P(C_2) = 61 / (79 + 61) = 0.44$

IAIR Est. 1986

用特征描述“某类会产生怎样的样本”

$P(x|C_1) = ?$ $P($  $|\text{Water}) = ?$

Each Pokémon is represented as a vector by its attribute. → feature

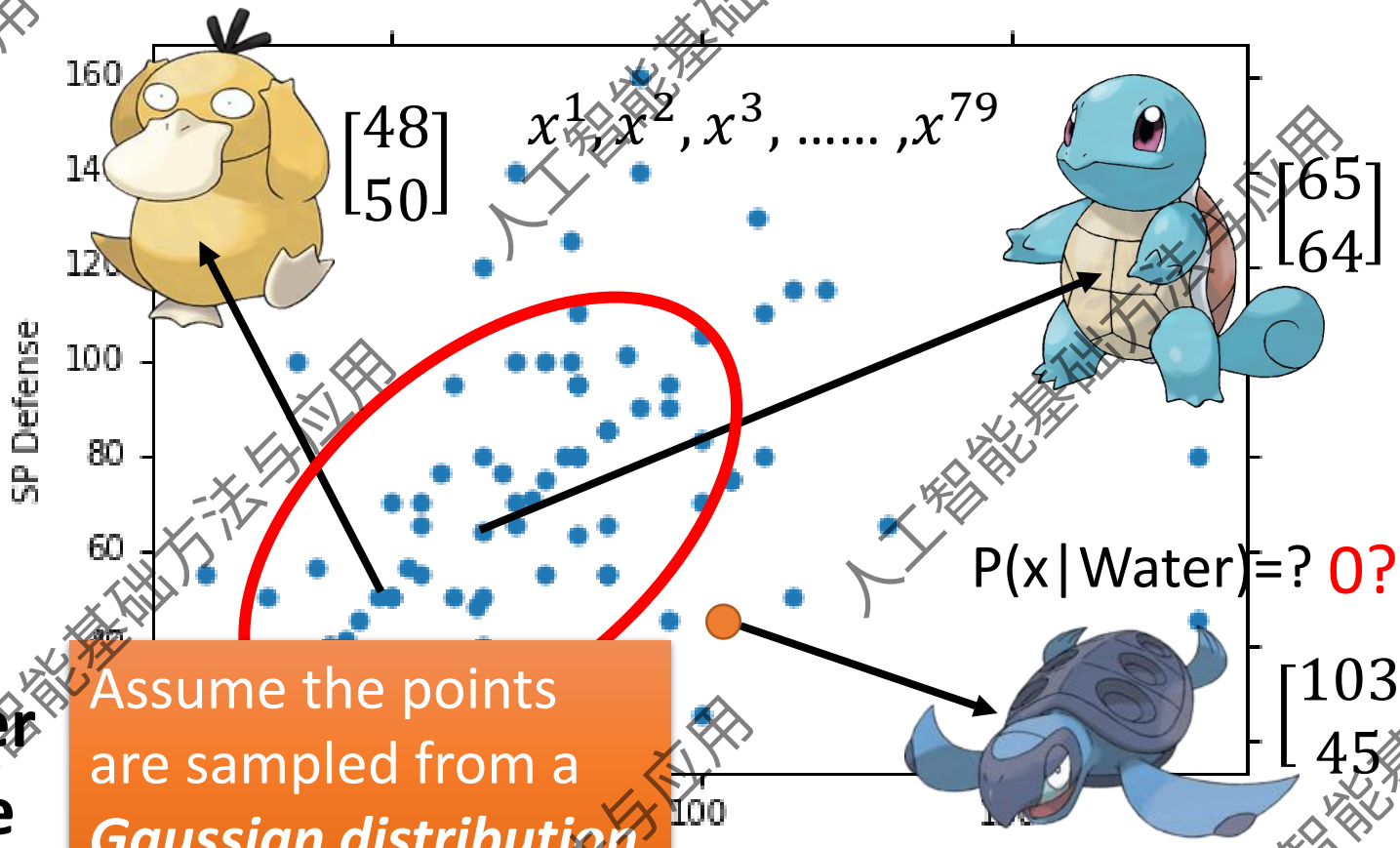


Water Type

人工智能基础方法与应用

从类别到特征的概率

- Considering **Defense** and **SP Defense**



Water Type

Assume the points are sampled from a ***Gaussian distribution***.

Gaussian Distribution

- "**大多数集中在中间，越极端越少**"的数据分布

Normal Distribution Formula

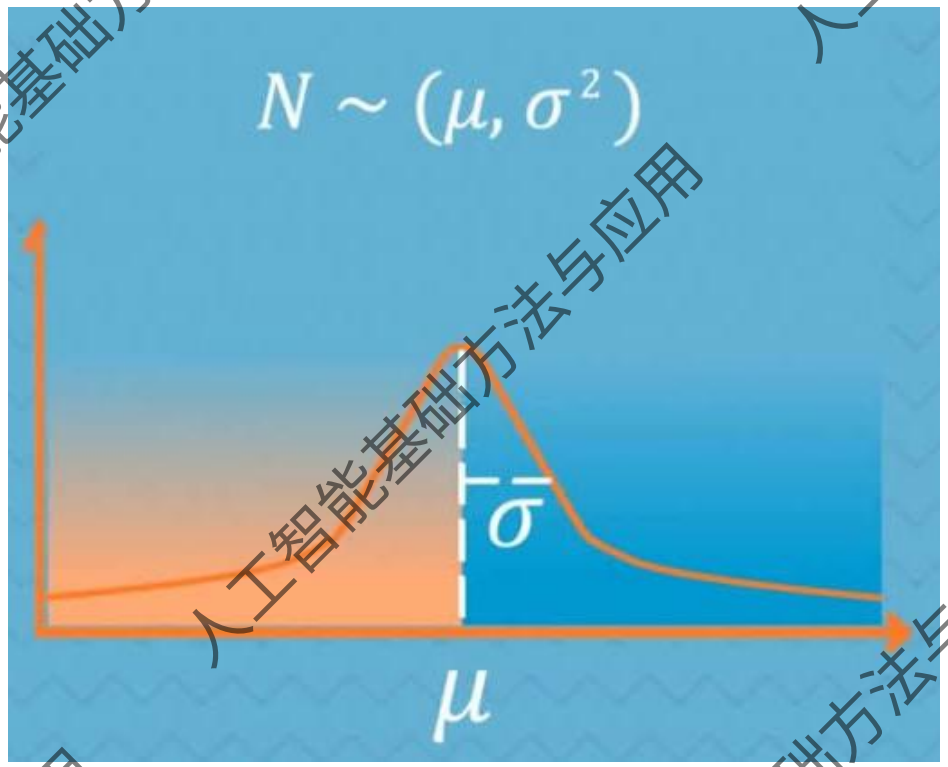
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$

$\mu =$ mean of x

$\sigma =$ standard deviation of x

$\pi \approx 3.14159\ldots$

$e \approx 2.71828\ldots$



人工智能基础方法与应用

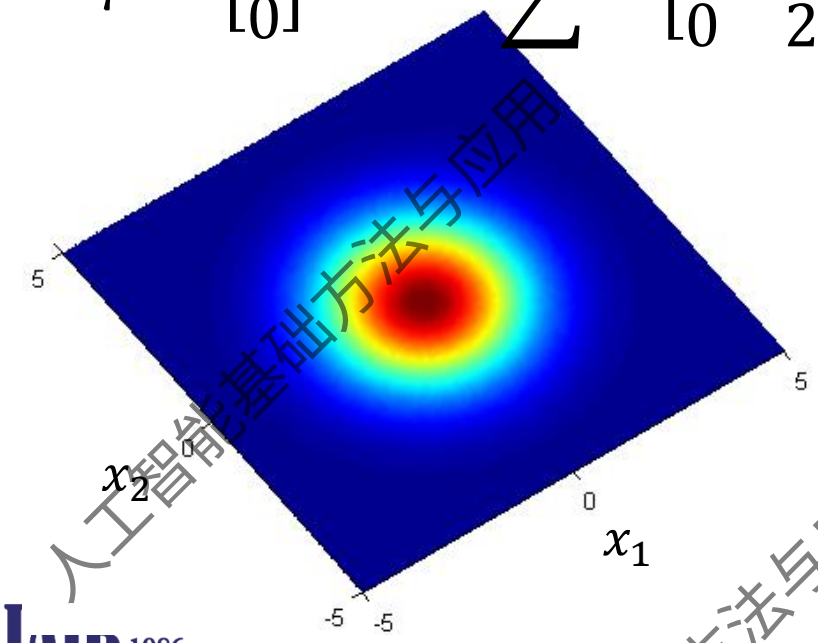
Gaussian Distribution

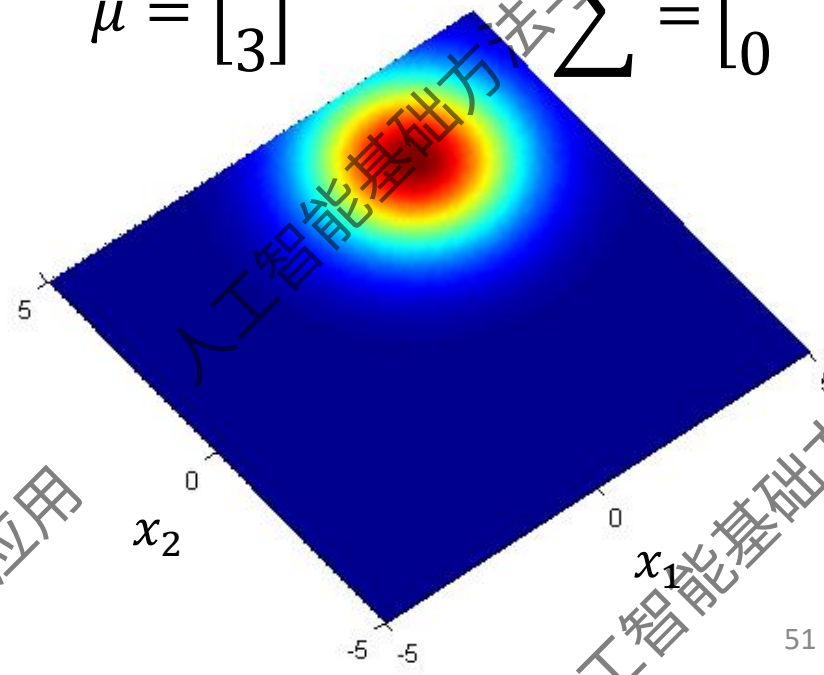
$$f_{\mu,\Sigma}(x) = \frac{1}{(2\pi)^{D/2}} \frac{1}{|\Sigma|^{1/2}} exp\left\{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right\}$$

Input: vector x, output: probability of sampling x

The shape of the function determines by **mean $\boldsymbol{\mu}$** and **covariance matrix $\boldsymbol{\Sigma}$**

$$\mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \qquad \sum = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \qquad \mu = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \qquad \sum = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$





人工智能基础方法与应用

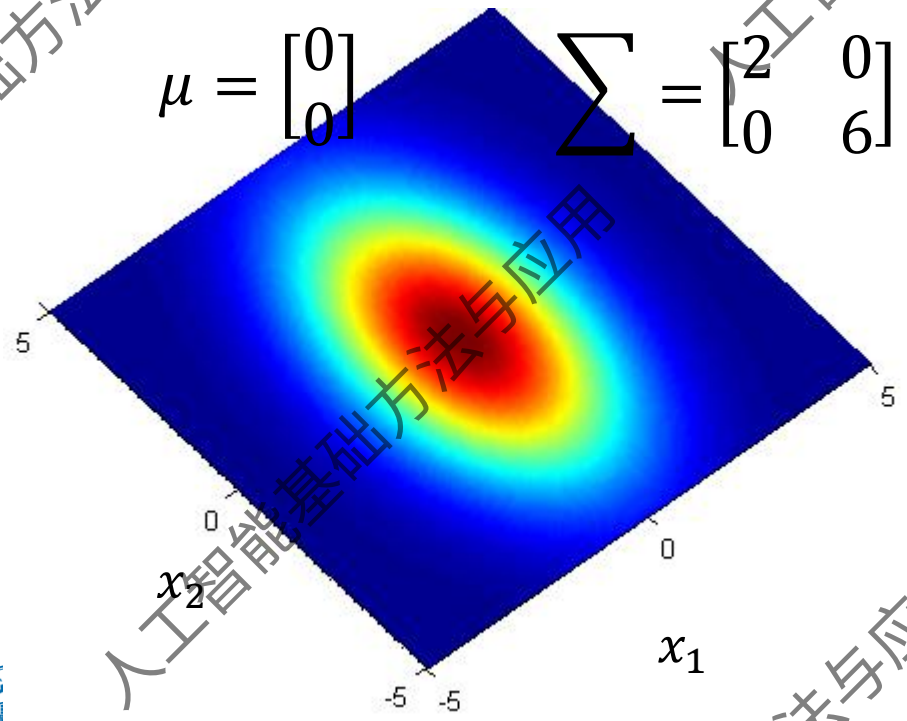
Gaussian Distribution

$$f_{\mu,\Sigma}(x) = \frac{1}{(2\pi)^{D/2}} \frac{1}{|\Sigma|^{1/2}} exp\left\{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right\}$$

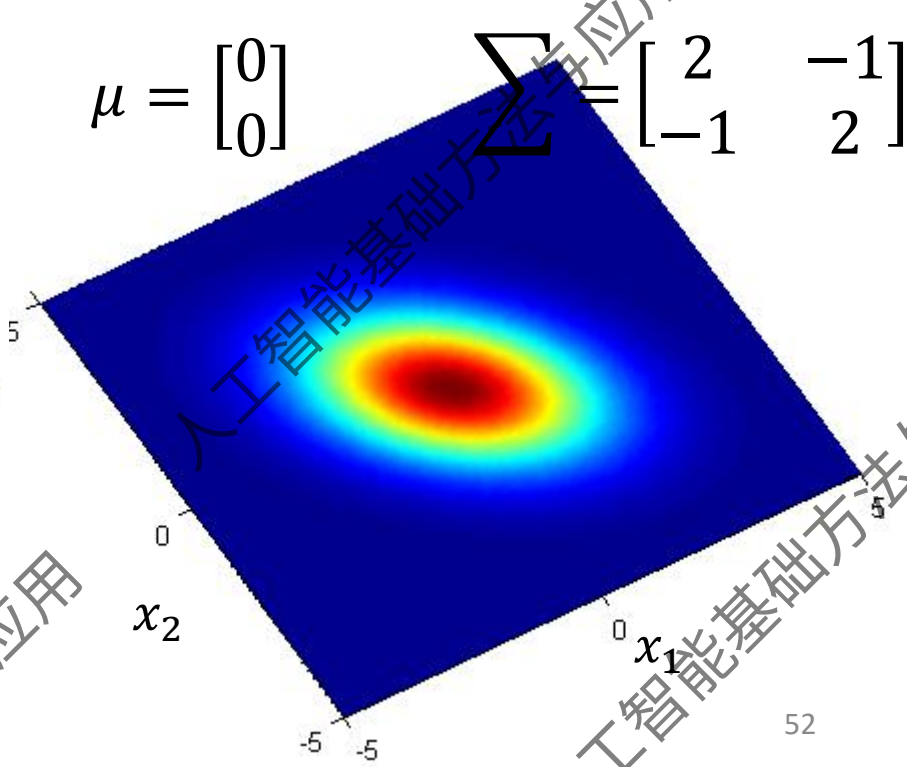
Input: vector x, output: probability of sampling x

The shape of the function determines by **mean $\boldsymbol{\mu}$** and **covariance matrix $\boldsymbol{\Sigma}$**

$$\mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \sum = \begin{bmatrix} 2 & 0 \\ 0 & 6 \end{bmatrix}$$



$$\mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \sum = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

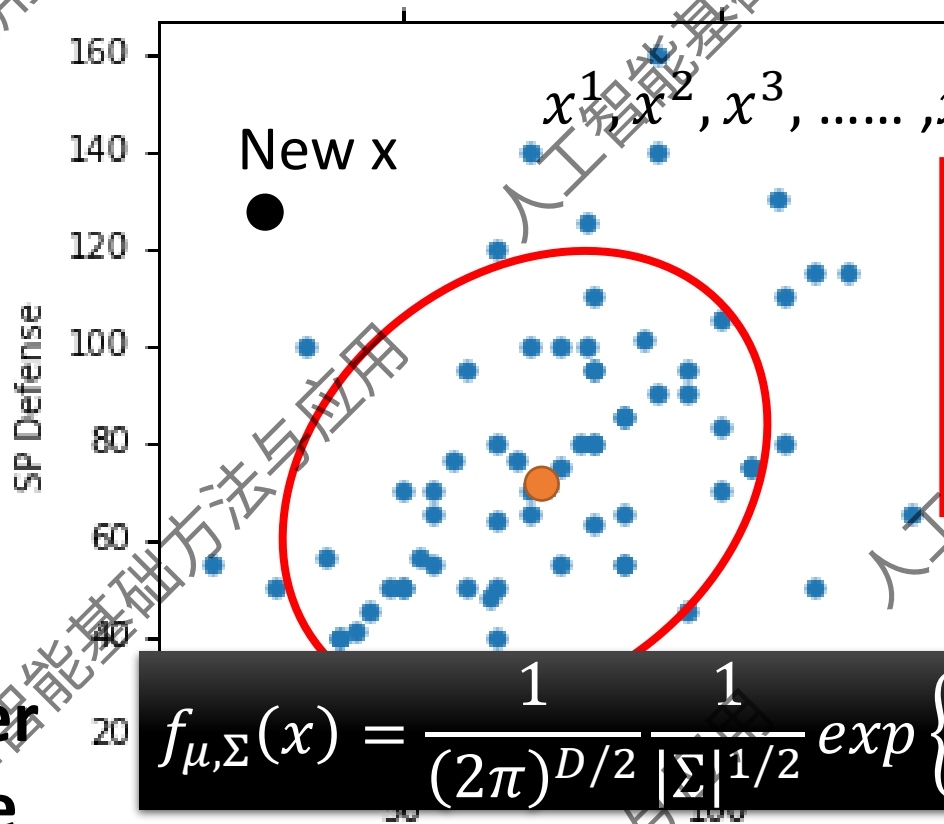


从样本反推最合适的分布

Assume the points are sampled from a Gaussian distribution

Find the Gaussian distribution behind them

Probability for new points



$$\mu = \begin{bmatrix} 75.0 \\ 71.3 \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} 874 & 327 \\ 327 & 929 \end{bmatrix}$$

How to find them?

$$f_{\mu,\Sigma}(x) = \frac{1}{(2\pi)^{D/2}} \frac{1}{|\Sigma|^{1/2}} exp\left\{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right\}$$

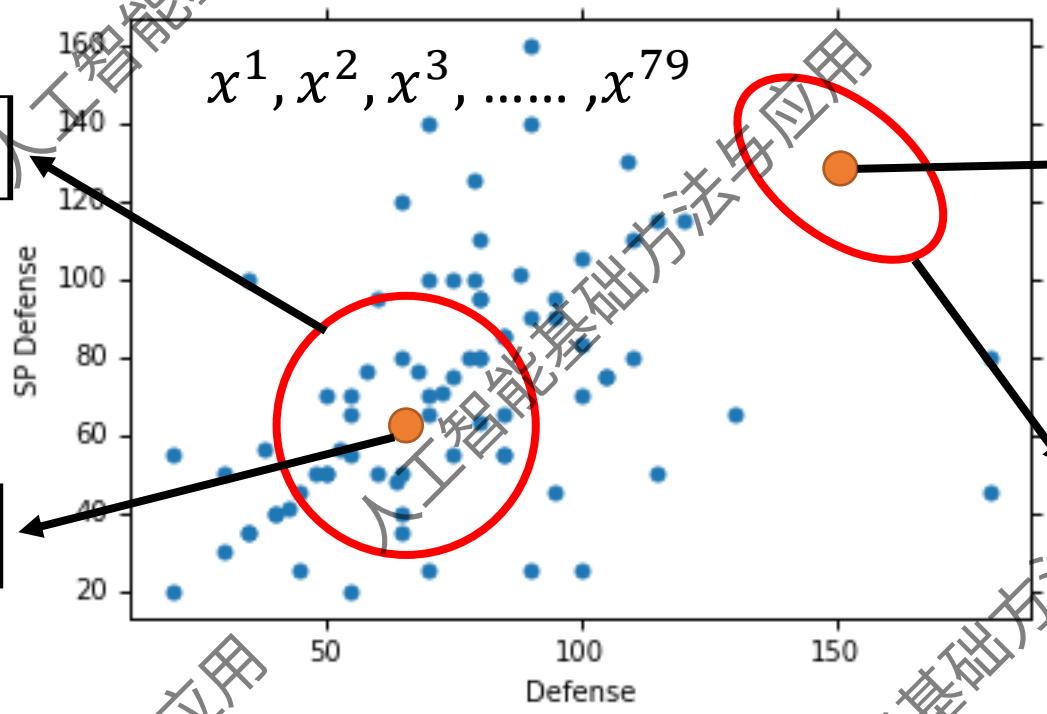
Water Type

人工智能基础方法与应用

Maximum Likelihood

$$f_{\mu,\Sigma}(x) = \frac{1}{(2\pi)^{D/2}} \frac{1}{|\Sigma|^{1/2}} exp\left\{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right\}$$

$$\sum = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$$



$$\mu = \begin{bmatrix} 150 \\ 140 \end{bmatrix}$$

$$\mu = \begin{bmatrix} 75 \\ 75 \end{bmatrix}$$

$$\sum = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

The Gaussian with any mean μ and covariance matrix Σ can generate these points. → Different Likelihood

Likelihood of a Gaussian with mean μ and covariance matrix Σ

= the probability of the Gaussian samples $x^1, x^2, x^3, \ldots\ldots, x^{79}$

$$L(\mu, \Sigma) = f_{\mu,\Sigma}(x^1) f_{\mu,\Sigma}(x^2) f_{\mu,\Sigma}(x^3) \ldots\ldots f_{\mu,\Sigma}(x^{79})$$

人工智能基础方法与应用

Maximum Likelihood

We have the "Water" type Pokémons: $x^1, x^2, x^3, \ldots\ldots, x^{79}$

We assume $x^1, x^2, x^3, \ldots\ldots, x^{79}$ generate from the Gaussian (μ^*, Σ^*) with the ***maximum likelihood***

$$L(\mu, \Sigma) = f_{\mu,\Sigma}(x^1) f_{\mu,\Sigma}(x^2) f_{\mu,\Sigma}(x^3) \ldots\ldots f_{\mu,\Sigma}(x^{79})$$

$$f_{\mu,\Sigma}(x) = \frac{1}{(2\pi)^{D/2}} \frac{1}{|\Sigma|^{1/2}} exp\left\{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right\}$$

$$\mu^*, \Sigma^* = arg \max_{\mu,\Sigma} L(\mu, \Sigma)$$

$$\mu^* = \frac{1}{79}\sum_{n=1}^{79} x^n$$

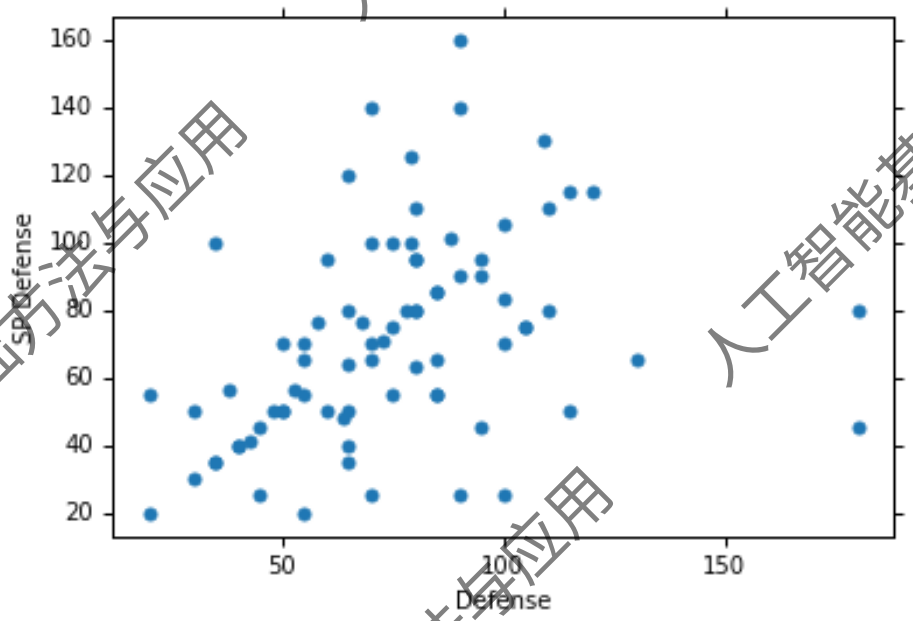
average

$$\Sigma^* = \frac{1}{79}\sum_{n=1}^{79} (x^n - \mu^*)(x^n - \mu^*)^T$$

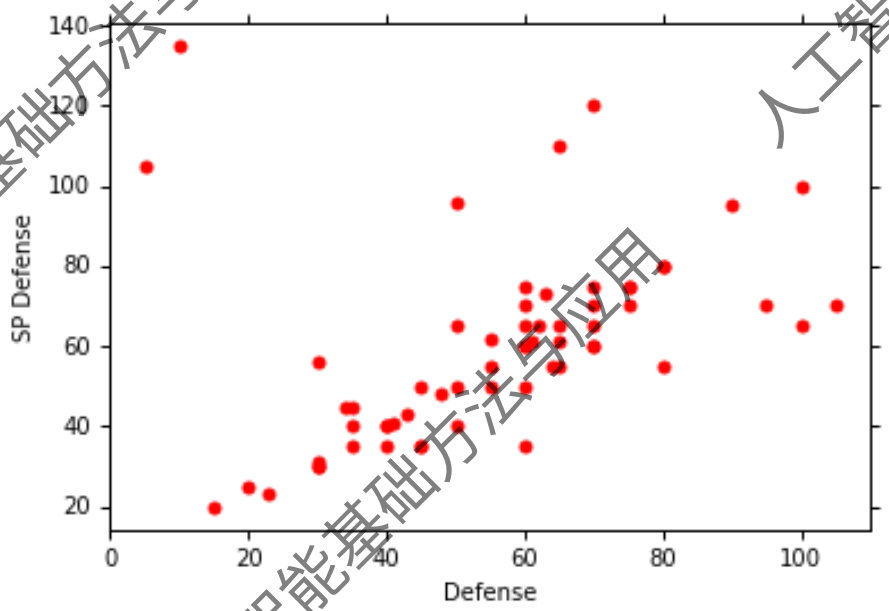
人工智能基础方法与应用

Maximum Likelihood

Class 1: Water



Class 2: Normal



$$\mu^1 = \begin{bmatrix} 75.0 \\ 71.3 \end{bmatrix} \quad \Sigma^1 = \begin{bmatrix} 874 & 327 \\ 327 & 929 \end{bmatrix}$$

$$\mu^2 = \begin{bmatrix} 55.6 \\ 59.8 \end{bmatrix} \quad \Sigma^2 = \begin{bmatrix} 847 & 422 \\ 422 & 685 \end{bmatrix}$$

人工智能基础方法与应用

Now we can do classification ☺

$$f_{\mu^1,\Sigma^1}(x) = \frac{1}{(2\pi)^{D/2}} \frac{1}{|\Sigma^1|^{1/2}} exp\left\{-\frac{1}{2}(x-\mu^1)^T(\Sigma^1)^{-1}(x-\mu^1)\right\}$$

$$\mu^1 = \begin{bmatrix}75.0\\71.3\end{bmatrix} \quad \Sigma^1 = \begin{bmatrix}874 & 327\\327 & 929\end{bmatrix}$$

P(C1)
= 79 / (79 + 61) =0.56

$$P(C_1|x) = \frac{P(x|C_1)P(C_1)}{P(x|C_1)P(C_1) + P(x|C_2)P(C_2)}$$

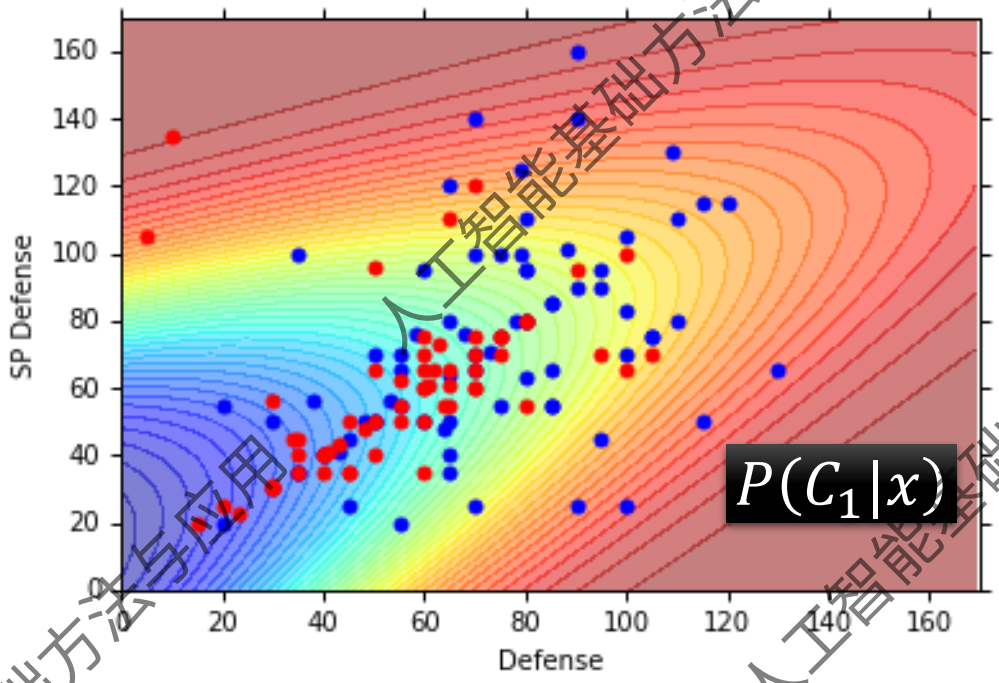
$$f_{\mu^2,\Sigma^2}(x) = \frac{1}{(2\pi)^{D/2}} \frac{1}{|\Sigma^2|^{1/2}} exp\left\{-\frac{1}{2}(x-\mu^2)^T(\Sigma^2)^{-1}(x-\mu^2)\right\}$$

$$\mu^2 = \begin{bmatrix}55.6\\59.8\end{bmatrix} \quad \Sigma^2 = \begin{bmatrix}847 & 422\\422 & 685\end{bmatrix}$$

P(C2)
= 61 / (79 + 61)
=0.44

If $P(C_1|x) > 0.5$ ➡ x belongs to class 1 (Water)

人工智能基础方法与应用



Blue points: C_1 (Water), Red points: C_2 (Normal)

How's the results?

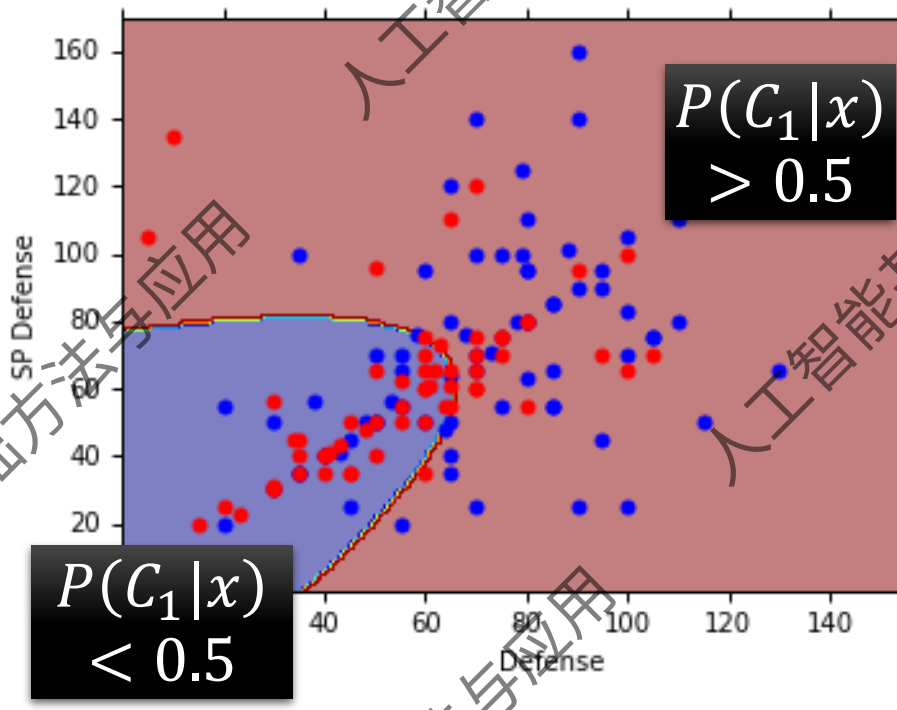
Testing data: 47% accuracy

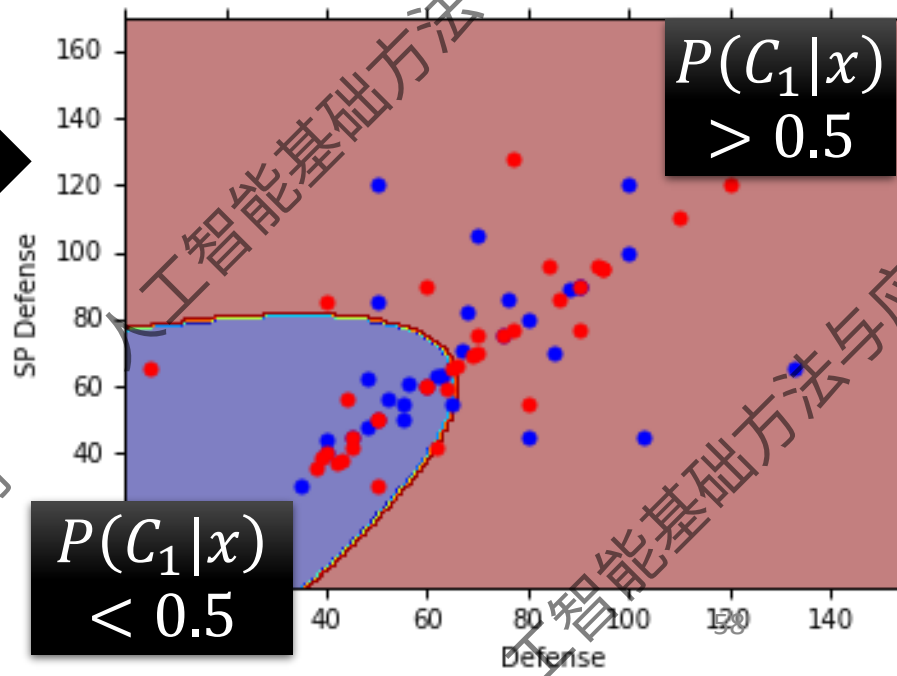
All: total, hp, att, sp att, de, sp de, speed (7 features)

μ^1, μ^2: 7-dim vector

Σ^1, Σ^2: 7 x 7 matrices

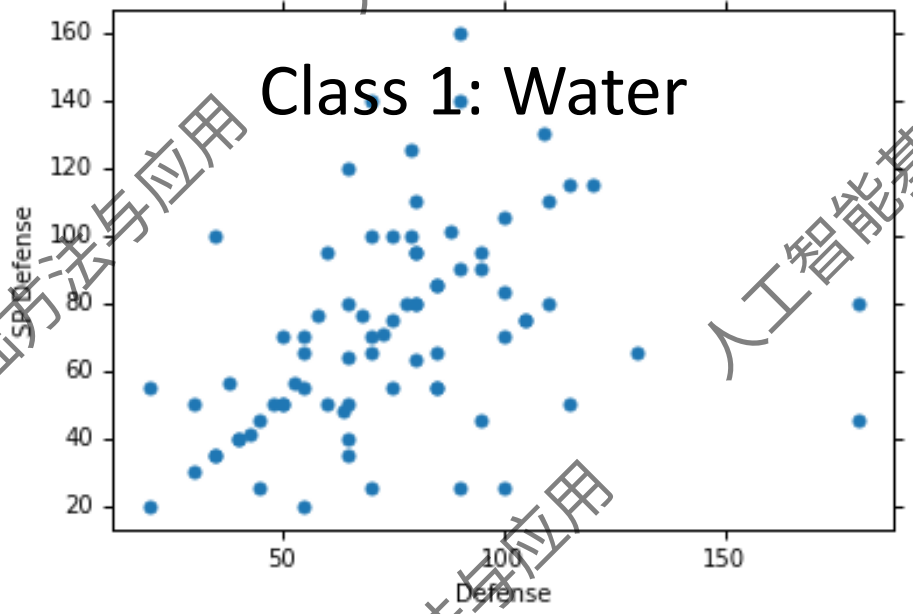
54% accuracy …☹

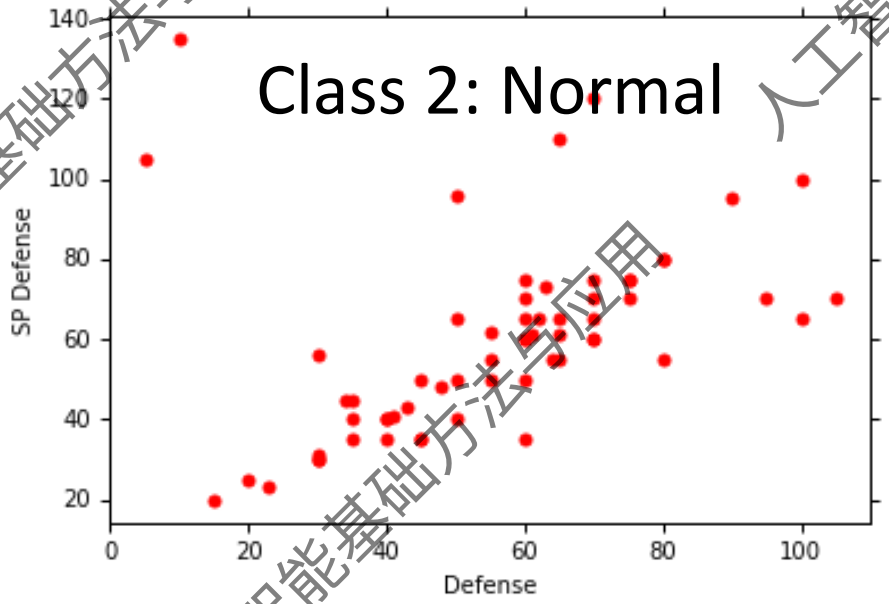




人工智能基础方法与应用

共享协方差以减少模型复杂度





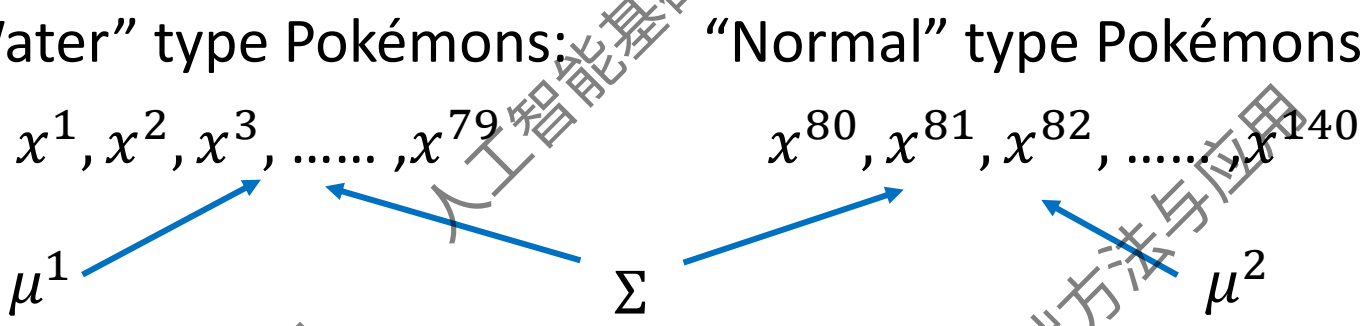
$$\mu^1 = \begin{bmatrix} 75.0 \\ 71.3 \end{bmatrix} \quad \Sigma^1 = \begin{bmatrix} 874 & 327 \\ 327 & 929 \end{bmatrix} \qquad \mu^2 = \begin{bmatrix} 55.6 \\ 59.8 \end{bmatrix} \quad \Sigma^2 = \begin{bmatrix} 847 & 422 \\ 422 & 685 \end{bmatrix}$$

The same Σ

Less parameters

Modifying Model

- Maximum likelihood



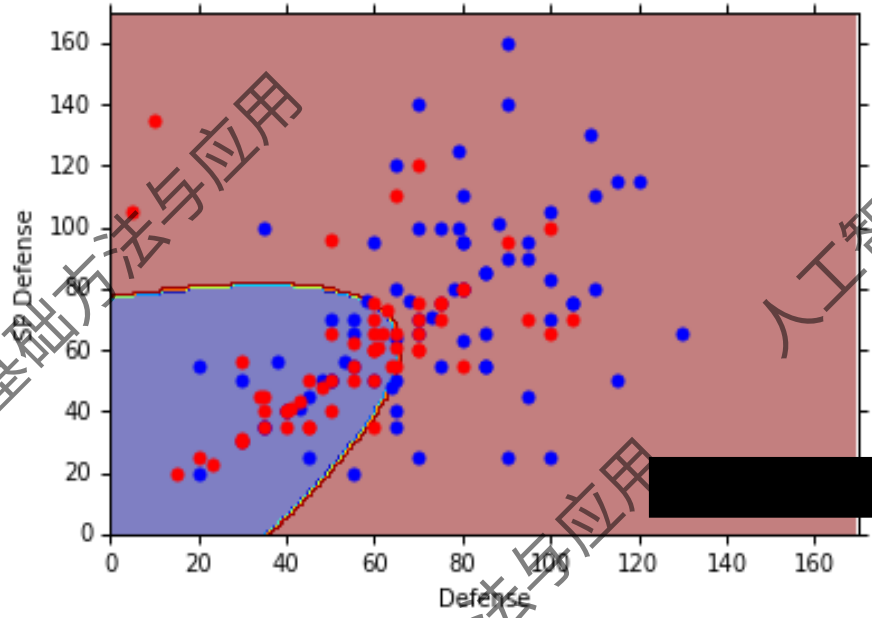
Find μ^1, μ^2, Σ maximizing the likelihood $L(\mu^1,\mu^2,\Sigma)$

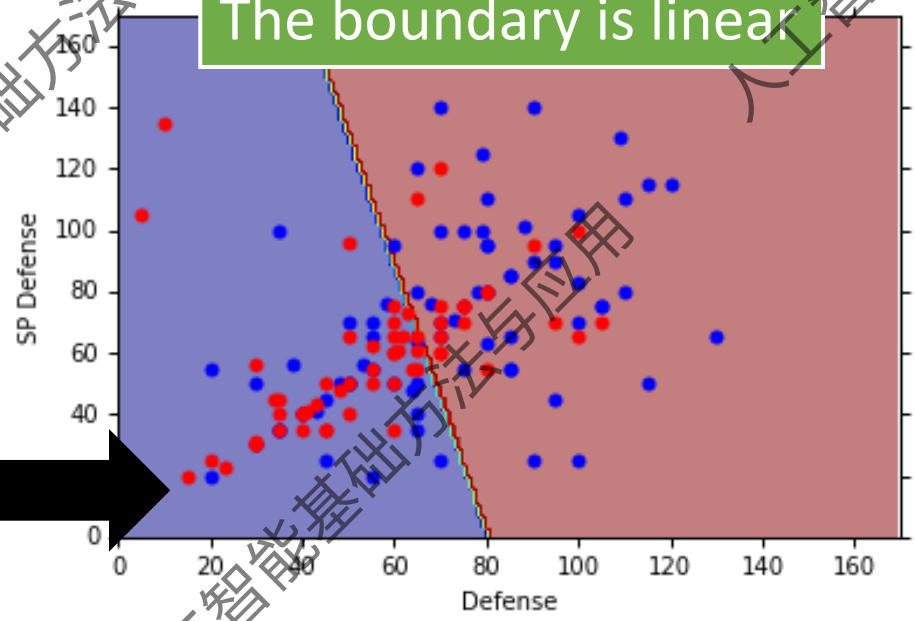
$$L(\mu^1,\mu^2,\Sigma) = f_{\mu^1,\Sigma}(x^1) f_{\mu^1,\Sigma}(x^2) \cdots f_{\mu^1,\Sigma}(x^{79}) \times f_{\mu^2,\Sigma}(x^{80}) f_{\mu^2,\Sigma}(x^{81}) \cdots f_{\mu^2,\Sigma}(x^{140})$$

μ^1 and μ^2 is the same $\quad \Sigma = \frac{79}{140}\Sigma^1 + \frac{61}{140}\Sigma^2$

人工智能基础方法与应用

更简单的模型也可能更准确





The same covariance matrix

All: total, hp, att, sp att, de, sp de, speed

54% accuracy → 73% accuracy

人工智能基础方法与应用

Three Steps

- Function Set (Model):

x ➡

$$P(C_1|x) = \frac{P(x|C_1)P(C_1)}{P(x|C_1)P(C_1) + P(x|C_2)P(C_2)}$$

If $P(C_1|x) > 0.5$, output: class 1

Otherwise, output: class 2

- Goodness of a function:
  - The mean μ and covariance Σ that maximizing the likelihood (the probability of generating data)
- Find the best function: easy

人工智能基础方法与应用

概率分布的选择

- You can always use the distribution you like ☺

$$P(x|C_1) = P(x_1|C_1)\ P(x_2|C_1) \quad \cdots\cdots \quad P(x_k|C_1) \quad \cdots\cdots$$

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_k \\ \vdots \\ x_K \end{bmatrix}$$

1-D Gaussian

For binary features, you may assume they are from Bernoulli distributions.

If you assume all the dimensions are independent, then you are using *Naive Bayes Classifier*.

课堂作业

- 请利用**线性回归**方法，根据温度、湿度、风速、降雨量、降雪量、季节和节假日等特征，预测共享单车租赁数量。数据来源于真实的 Seoul Bike Sharing Demand 数据集
- **数据说明（微信群内提供：共享单车回归作业数据.zip）**
  本作业提供训练集：100 条数据，测试集：60 条数据
- **作业要求**

1. 建立线性回归模型：$y = b + \sum w_i x_i$
2. 使用 100 条训练数据训练线性回归模型，说明数据处理和模型求解过程，并给出模型学习得到的参数b和w_i
3. 分析线性回归模型在训练集和测试集上的表现
4. 提交时限：两周之内
5. 提交PDF格式电子文档，以***姓名-学号-共享单车回归作业***命名，发送至助教邮箱：Email：liushuaiyu@stu.xjtu.edu.cn

完整数据集：https://archive.ics.uci.edu/dataset/560/seoul+bike+sharing+demand

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人工智能基础方法与应用



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2023.06

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